The Fable Computer: A Room-Temperature Terahertz Half Adder on a Regenerative Graphene-Plasmon Logic Fabric

Concept, Architecture, and Reduced-Order Feasibility

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The machine described here is named the Fable Computer: a half adder — two input bits in, Sum and Carry out — realized on a clocked, regenerative graphene-plasmon logic fabric. Throughout this manuscript, "the fabric" names the technology and "the half-adder block" names the demonstrator built from it.

Authorship and Status Disclaimer. This manuscript is derived from the author's original paper [1]. Generation of content involved the use of an AI, Anthropic's Fable 5, at the author's direction, including an AI-orchestrated literature-verification and consistency audit (June 2026) in which every reference was checked against its primary source and every quantitative claim independently re-derived. The account of the community notes promoted from 31 July through 17 October 2026, and a re-check of external facts against primary sources on 24–25 October 2026, were prepared in a maintainer-operated Claude Code session in which Claude Fable 5.1 orchestrated Claude Opus 5.5 agents for mapping, two-lens adversarial verification, drafting and checking; the community notes the text relies on were themselves produced and assessed by the project's disclosed agent pipeline, each note's header giving its promotion route and, where the note records them, its executing models. The author declares a lack of formal expertise in graphene-plasmonic device fabrication and full-wave electromagnetic simulation, and the document has not received independent expert review. The architectural and quantitative claims herein must be treated solely as a starting point for discussion, not as peer-reviewed engineering guidance. Formal expert review is required prior to any practical implementation.

Abstract

We present a machine model for room-temperature terahertz logic. The bit is a gate-screened (acoustic) graphene plasmon. Every logic gate is the same physical object — a resonant Dyakonov–Shur (DS) cell biased below its self-oscillation threshold and operated as a regenerative amplifier — and the system clock is a phase-stable pulse train launched at the fabric edge by a microcomb-photomixing chain. In the design model of Section 5, logic levels are restored at every gate by saturating regenerative gain, and inversion is a bias choice. The released model qualifies both (Section 5). Its cell has no output plateau (demonstrated); its soft compression re-clamps a loaded fan-out-2 cascade near a 2 % drive rather than at a rail (in-model); two summed '1's read only 2.6–3.3 dB above one (demonstrated); and no logic input depletes it (in-model). Short of an external bias swing, the only amplitude-encoded inverter it admits is destructive interference with the clock pulse, inside a 10.1–12.3° phase window that narrows as the solver grid is refined

(demonstrated). Under a phase encoding, NOT becomes a half-period delay, and a $2f_0$ pump restores phase, but only in narrow windows (in-model). Two constraints are locked from the outset: no cryogenics, and chip temperature ≤ 80 °C.

The demonstrator is a half adder: five regenerative cells computing Sum and Carry at one addition per 4-ps slot (2.5×10^{11} additions/s, wave-pipelined) as the design baseline, at ∼0.4 fJ per addition on the fabric side at that slot (the figure scales with the slot length, Section 8.1), in roughly 12 × 16 μm. In-model, fan-out-2 logic does not close at the 4-ps slot once cavity ring-down is booked; with a physical launch confined to its slot and gated by the data it closes at ≈0.07 THz (69 GHz, band 63–74 GHz; neither confirmed nor moved when the bias is set as a ratio to the solver's own threshold, Section 6.2). A five-pass reduced-order feasibility chain converges on one self-consistent operating point — n = 10^{12} cm^{-2}, cell length L ≈ 576 nm tuned to 1 THz, drift bias parked at 0.7 of the self-oscillation threshold (the tracked ratio M/M_th = 0.7, Section 5) — at which a single cell delivers ∼+8 dB net gain per pulse as measured by the released solver's streaming per-slot-peak metric (a +9.7 dB continuous-wave regenerative gain; the per-pulse figure carries the calibration budget quantified in Section 7.3, and a design-length rerun measures +9.2 dB, confirming the pre-registered correction) and, on the analytic cascade budget of Section 7.4, two cascaded cells hold +8.4–9.3 dB per cell through junctions as lossy as −6 dB, all in-model (a term defined in Section 7: established within the reduced-order chain, not measured). Every gain figure quoted here is a hydrodynamic-tier figure, and each carries a signed kinetic correction band: −0.95 dB per cell at 353 K at the tracked operating point, band [−1.5, −0.3] dB, in-model within the next tier's closure (at 300 K the same closure gives −1.27 dB, band [−1.6, −0.5] dB; whether a 353 K threshold exists to track at the kinetic tier is open, Section 7.5) — plus a separate and still-open ±1 dB allowance for the kinetics of the clamped contacts — sign-open at the drain, loss-definite at the source, and potentially larger than ±1 dB for a diffuse contact (Section 7.5). The correction's sharpest consequence is thermal rather than logical: the 353 K margin of Section 8.2 reads ≈ 4 % rather than 23 % within the closure once the shifted threshold is paid for in bias current, and in every computed row of graphene's own electron fluid that has a threshold, tracking it at that fluid's own Joule dissipation overruns the 55 K budget (in-model, Section 8.2).

Three items remain open and are stated as such: a Boltzmann–Maxwell solve owning the kinetic gain correction — and, first, whether the cell has a 353 K threshold at all — electromagnetic coupling, and absolute calibration; the photonic synthesis of the pulsed clock, which extends a demonstrated continuous-wave chain; and a bench experiment, for which a five-gate pass/fail protocol with pre-registered thresholds is specified. The most informative single outcome, pass or fail, is the first cascade of two room-temperature plasmonic gain cells.

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1. Introduction: the problem of the picosecond bit

A graphene plasmon is an attractive carrier for logic. It is programmable — a gate voltage sets, point by point, whether a region of the sheet supports the wave at all — and at terahertz frequencies the gate-screened (acoustic) mode confines the field to deep-sub-

wavelength scales on a chip that needs no cryostat. Whether the bit is a genuine wave or an overdamped relaxation is decided by one number, the quality factor $Q = \omega\tau$, where τ is the plasmon's momentum-relaxation lifetime. With the best graphene encapsulated in hexagonal boron nitride (hBN) — $\tau \approx 1$ ps at 300 K, a best-case value examined in Section 4 — Q reaches ≈ 6 at 1 THz.

The choice of ~1 THz as the carrier is a systems compromise, not a materials ceiling. Below ~0.5 THz two floors intrude: the quarter-wave cell grows well past its ~500-nm lithography pitch, surrendering the footprint advantage that justifies plasmons in the first place, and the carrier falls to within a few multiples of the 0.2–0.5 THz symbol rate, leaving too few carrier cycles to form a clean clock pulse; far above 1 THz the drive and read-out chain (photomixer power, demonstrated stripline bandwidth, Section 3) runs out first. The intrinsic loss channels of the stack — acoustic phonons and residual impurities — vary smoothly across this band; graphene's optical phonons (≈ 47 THz) and hBN's Reststrahlen bands (≈ 23–48 THz) lie more than an order of magnitude higher and play no role at 1 THz.

The same Q states the central obstacle. A lifetime of a picosecond means the wave loses ~4.3 dB of power per plasmon wavelength at 300 K (5.1 dB at the 80 °C cap): launched passively, a bit decays below any usable read threshold within a handful of gates. A machine — as opposed to a one-shot demonstration — therefore needs four things a bare plasmonic circuit does not have: distributed gain that at least cancels the loss at every gate; level restoration, so that decisions made early in a cascade do not erode; a clock distributable across millimetres at femtosecond-class phase; and memory, since nothing plasmonic survives between operations.

Each requirement is assigned to hardware that can plausibly deliver it. Gain is electrical. Each gate is a tiny plasmon cavity carrying a DC drift current — the configuration in which Dyakonov and Shur predicted that a gated transistor channel with asymmetric terminations goes unstable and self-oscillates at its plasma resonance [11]. Biased just below that self-oscillation threshold, the cavity acts as a regenerative amplifier: an injected pulse recirculates through the gain region many times before it leaves, and in the design model the saturating response doubles as the logic nonlinearity (Section 5, which also records what the released model executes: a cell that compresses rather than thresholds). The experimental basis is deliberately stated with care: room-temperature, current-driven amplification of terahertz radiation by graphene plasmons has been demonstrated once, by Boubanga-Tombet et al. [2] — but its authors explicitly attribute the effect to a dissipative, phase-shift mechanism without any plasma instability, not to the DS mechanism, and follow-up theory offers a third reading [22]. Reference [2] therefore proves that DC current can pump energy into graphene plasmons at 300 K; it does not demonstrate the DS-cavity gain this design assumes, which remains experimentally unconfirmed (Section 2). Optically pumped gain, for its part, cannot be sustained beyond a sub-picosecond transient [3].

Timing is optical: the bias plane's RC corner (~1 MHz at chip scale, ~10 GHz even per 10-μm segment) is orders of magnitude below 1 THz, so no electrically modulated clock is physically available; a microcomb-photomixing chain at the fabric edge launches the clock as a pulse train (Section 3.2 — with the pulse synthesis itself an extension of demonstrated hardware, flagged there). Restoration is intrinsic in the design model: every gate is the

same gain cell, and its saturating transfer re-clamps the levels. The released cell's soft compression re-clamps a loaded cascade's '1' near a 2 % drive rather than at a rail, but it does not meet the regeneration condition (in-model, Section 5). Memory is electronic: bits persist as gate charge, not as plasmons (Section 6.4).

The machine built from these answers is deliberately the smallest complete one: a half adder — five regenerative cells computing Sum and Carry — specified to the level of cell map, timing, gain budget, and energy in Section 6.5, and feasibility-checked by the five-pass model chain of Section 7, with Section 5 recording what the released cell itself executes. Section 8 gives the energy and thermal accounting; Section 9 collects device-implementation considerations; Section 10 reduces everything still unproven to five bench gates — pre-registered pass/fail milestones for the experiment, "gate" in the protocol sense, not logic gates — and one remaining solve.

2. Related work and positioning

Current-driven plasmon gain. The experimental anchor is the single demonstration of room-temperature, DC-current-driven terahertz plasmonic amplification, in asymmetric dual-grating-gate graphene structures: ~9 % radiation gain (≈ 0.4 dB), probed by pulsed time-domain spectroscopy [2]. Its authors exclude the Dyakonov–Shur instability as the explanation (resonance width does not narrow with current) and propose a dissipative phase-shift mechanism; later theory re-reads the same data as an amplified-mode-switching effect [22]. A primary-source re-check of the 2025–26 literature (October 2026) found no experiment claiming net terahertz gain, emission or DS self-oscillation from drifting electrons in graphene or any other two-dimensional channel (external). The nearest results are an earlier (2024) current-driven amplification of terahertz sound — acoustic phonons, not radiation, at cryogenic temperature [43] — and a current-induced extra damping of mid-infrared plasmons traveling against the current at 170 K, which its authors attribute to Cherenkov emission of acoustic phonons [44]. The DS instability itself [11] — asymmetric boundary conditions (AC-shorted gate-to-channel voltage at source, AC-open current at drain) driving a cavity instability — has a substantial theory literature for graphene, including hydrodynamic treatments with radiative loading [19], giant-gain grating-gate variants in simulation [21], and amplified-reflection instabilities below the Cherenkov threshold [20]. A kinetic Boltzmann–Maxwell treatment of drift-driven gain in gated graphene has recently appeared [18]. What does not exist anywhere in the literature, to the author's knowledge, is this design's elemental operation: sub-threshold regenerative amplification of injected plasmon pulses by a DS cavity. The closest precedents are regenerative resonance narrowing below threshold, demonstrated for detection [23], and the amplified drain-boundary reflection that is the textbook gain event of the DS mechanism [20]. This gap is exactly what bench gates G1–G4 of Section 10 test.

Drift nonreciprocity. A separate literature establishes how a DC drift bias breaks reciprocity for graphene plasmons: the co-drift (downstream) wave is dragged to longer wavelength and lower spatial damping, the counter-drift wave to stronger damping — measured directly as plasmonic Fizeau drag [24, 25] and analyzed for nonreciprocal devices [26, 27]. Section 6.1 builds the fabric's routing on this sign: signals travel with the electron drift.

Competing room-temperature logic platforms. Table 1 positions the proposed fabric against the technologies a referee would name. The honest summary: CMOS wins on every axis except clock rate; superconducting single-flux-quantum logic (RSFQ, with its adiabatic AQFP variant) is excluded here by the no-cryogenics constraint; THz-capable electronic amplifiers (indium-phosphide high-electron-mobility transistors, InP HEMT: 9 dB at 1 THz [28]) and oscillators (resonant-tunneling diodes, RTD, to 1.92 THz [29]) exist but lack a restoring, cascadable logic family at those frequencies; graphene THz nonlinearities are strong [30] but unclocked. The niche claimed by this design is specific: restored, cascadable, wave-pipelined logic without cryogenics, from a clock chain specified at 0.2–0.5 THz (with ring-down booked and a physical, slot-confined, data-gated launch, fan-out-2 logic closes in-model at ≈0.07 THz, Section 6.2) — a niche currently occupied by nothing, including this design, whose own gain cell is unproven. The comparison is offered as motivation, not as a claim of superiority.

Technology	**Clock**	**Energy/op**	**Restoring logic?**	**Cryo-free?**	**Status**
CMOS (IRDS 2023 [17])	~GHz	~aJ–fJ	yes	yes	manufactured
RSFQ/AQFP supercond. [31]	10–100 GHz	~aJ	yes	no (4 K)	mature niche
InP HEMT THz amps [28]	to ~1 THz (analog)	—	no (amplifier)	yes	demonstrated
RTD oscillators/latches [29]	to ~2 THz (osc.)	~fJ	latch only	yes	demonstrated
Photonic logic [16]	10–100 GHz	~fJ system	weakly	yes	research
Spin-wave/magnonic [32]	~GHz	~fJ	weakly	yes	research
This fabric (proposed)	0.2–0.5 THz clock spec; fan-out-2 logic ≈0.07 THz in-model (Section 6.2)	~0.4 fJ fabric-side at the 4-ps baseline slot, growing with the slot (Section 8.1)	yes in the design model (regenerative); the released cell re-clamps levels by compression (in-model) but does not threshold (demonstrated), Section 5	yes (≤ 80 °C)	paper design; gain cell unproven

Table 1. *Room-temperature-relevant logic technologies. Energy figures are order-of-magnitude and not normalized for function; the fabric-side figure excludes the photonic drive chain, which dominates (Section 8.1).*

Subsystem precedents. Every subsystem of the fabric rests on a demonstrated technology; the fabric is the integration. Table 2 collects them, with the demonstration conditions stated — four of the five were demonstrated under conditions weaker than the fabric requires, and are flagged.

Subsystem	Demonstrated basis	Condition flags
Clock/data generation	microcomb-photomixing chain, chip-scale, phase-coherent [5]	CW two-tone at 560 GHz; pulsed-clock synthesis is a proposed extension (Section 3.2)
Clock-distribution phase control	optical interferometric locking, 9 mrad RMS residual [6]	demonstrated in a squeezed-light optical parametric amplifier at optical frequencies; cited as plausibility, not as a THz-clock demonstration
Launch, read-out, in-situ reference	monolithic generator/detector stripline circuit [7]	−3 dB bandwidth 440 GHz; extension toward 1 THz untested
Substrate and plasmon transport	sapphire / coplanar-waveguide (CPW) / hBN-graphene stack, on-chip THz plasmon wave packets [4]	demonstrated at 4 K; gated (screened) mode used a metal top gate
Free-space port (option)	graphene THz MIMO patch antenna, simulation-only, 3.75–6.46 THz [8]	no fabrication or measurement; cited as a design style
Per-cell gain mechanism	current-driven plasmonic THz amplification at 300 K [2]	~9 % radiative gain; mechanism attributed by its authors to dissipative phase shift, not DS

Table 2. *Subsystem precedents with demonstration-condition flags. The flags are part of the claim: where a precedent is weaker than the fabric's requirement, the gap is named here and re-appears among the open items of Sections 9–10.*

Relation to the author's previous paper [1]. This manuscript supersedes [1] in four ways: the carrier moves from 100 GHz to ~1 THz with the symbol rate carried by a comb-locked pulse train; the gain element changes from a generic amplifying section to a specific resonant DS cell operated regeneratively below threshold; the feasibility argument is rebuilt as the five-pass chain of Section 7 with every parameter exposed; and the claims are bounded by a pre-registered bench protocol (Section 10). The machine-learning design-optimisation thread of [1] is dropped entirely.

3. Physical platform and drive chain

3.1 Substrate, stack, and the programmable gate lattice

The chip is built on sapphire ($\varepsilon_r \approx 9.4$, $\tan\delta < 10^{-4}$) carrying a segmented Au gate lattice, hBN-encapsulated monolayer graphene with one-dimensional edge contacts [33], and a top cap; the cell cross-section is shown in Figure 3a. The stack follows the on-chip terahertz plasmon circuit of Yoshioka et al. [4], with two caveats stated plainly. First, that

demonstration was performed at 4 K; it is precedent for the launch-and-transport architecture, not for room-temperature plasmonics. Second, in [4] the gate-screened (acoustic) mode this design uses was obtained under a Ti/Au top gate, while the ZnO-capped device hosted the unscreened mode; the fabric's cells therefore assume metal top-gate segments, with hBN of thickness d = 10 nm as the gate dielectric, and ZnO remains an option only where a THz-transparent global window is wanted.

The graphene channel is dual-gated: a global back gate sets the working density ($n = 10^{12}$ cm^{-2}, $E_F \approx 117$ meV), while an electrode-mapping controller writes a per-segment voltage map onto the top-gate lattice. That density map is the program: it defines which cells are doped (active, amplifying) and which sit at the charge-neutrality point, CNP (dark, absorbing guard cells). The lattice pitch follows the gain cell of Section 5 — DS elements of $L \approx 576$ nm — comfortably within deep-UV lithography.

3.2 The photonic clock chain: microcomb → DFB pair → photomixer

The clock generator builds on the microcomb-photomixing chain demonstrated by Tokizane et al. [5] as a chip-scale, fiber-bonded, phase-coherent terahertz source: a SiN soliton microcomb (one endurance run sustained > 24 h at 1 W pump) whose adjacent lines injection-lock a pair of distributed-feedback (DFB) lasers (one IQ-modulated for data), recombined and photomixed in a uni-traveling-carrier photodiode (UTC-PD), achieving 112 Gbps wireless transmission at 560 GHz (Figure 1).

One distinction is load-bearing: the demonstrated chain is two-tone photomixing, and a two-tone photomixer emits a continuous-wave (CW) carrier — it cannot emit a pulse train of any kind. The fabric's clock — few-cycle pulses on a ~1 THz carrier at a 0.2–0.5 THz repetition rate — has a spectrum of four to six comb lines spaced at the symbol rate around 1 THz, and synthesizing it photonically requires (i) selecting that many phase-coherent comb lines (microcombs with 0.2–0.56 THz line spacing are demonstrated hardware [5, 34]), (ii) line-by-line amplitude and phase shaping, the established optical-arbitrary-waveform-generation technique [35], and (iii) a photomixer with adequate response across 0.5–1.5 THz [36]. Each ingredient exists; their combination at these frequencies does not yet. The pulsed clock is therefore classified as a proposed extension of demonstrated hardware, and joins the open-items list (Section 10). Which compact source class should supply the ≈1 THz carrier — this photomixing chain, a resonant-tunneling diode or a silicon harmonic source as master, with the plasmonic cell itself a candidate repeater at each re-sync station — whether the burst envelope is shaped at the source or cut from a continuous carrier downstream, and what distribution budget the clock needs, are open design questions, tracked as issue #139.

Available launch power bounds the budget. Demonstrated CW UTC-PD output falls from ~mW at 300 GHz to the microwatt class near 1 THz (2.9–10.9 μW at ≈1 THz across device classes, and 24 μW at 0.914 THz from a traveling-wave device [36, 37, 38]; a primary-source re-check in October 2026 found no 2020–26 UTC-PD result above these, external); at ~3 μW, one 4-ps clock slot carries ~10 aJ — comparable to the fabric's ~10 aJ/bit field energy (Section 8.1) before any coupling loss. The launch budget therefore closes only if the edge transducer from the photomixer's mode into the acoustic-plasmon mode is efficient;

that coupling efficiency is unquantified by any cited work and is named as an open item (Sections 9, 10).

Phase at the clock source is the easier half. Optical interferometric locks reach ~9 mrad RMS residuals [6] — 1.4 fs at 1 THz against the fabric's 42 fs (±15°) budget, a ~30-fold margin. The cited demonstration is from a squeezed-light parametric-amplifier system at optical frequencies, so it is invoked as evidence of the achievable class of optical phase control, not as a THz-clock demonstration. The cells draw on the same budget. In-model, a cell's own transfer phase moves 12–21° per 1 % of drive detuning at the carrier and 4–6° per 0.05·M_th of bias, so one cell spends the whole ±15° at about ±1 % of f_0 — 4–10 nm of cell length at the ratio bias of Section 5 (2.7–10 nm across configurations, Section 9.7) — or, in the released chain's shipped cavity, at +0.10 / −0.22·M_th of bias (notes/2026-10-13-phase-is-not-the-easier-half-for-the-cell.md; the dated notes cited by file path throughout this paper are the promoted notes of the community record, one Markdown file each in the notes/ directory of the repository, https://github.com/ryoji-info/FableComputer/tree/main/notes, where notes/INDEX.md lists what each settles and what later notes did to it, and each note's header gives its promotion route). In the clocked regime of Eq. (5), where every cell amplifies the clock pulse, that is a single-cell requirement on trim and bias tracking. If instead a two-input gate sums the coherent fields of a station-free cascade, a between-branch error must stay within ±0.32 … ±0.010 % of f_0 per cell over the 3–91 cells its two branches do not share in Part III's 92-cell-deep lane [45], and the ±15° is then not met as a fabric-level budget; read as envelope-coupled, phase does not enter the gate, and the question is open. Which regime and which reading the fabric is in are open; Section 5 records that the released cell supplies no threshold AND for a field sum, and a $2f_0$-pumped cell, which restores phase in a narrow window, would not accumulate the per-cell error (in-model composition, within its slip margin — 19.2° at knee-level swings under a 5 % pump, but ≈ 6° at the delivered '1' under a 2 % pump, below the single-cell ±15°; how detuning moves a pumped cell's own axis was not computed; notes/2026-10-17-a-pumped-cell-restores-phase-in-a-narrow-window.md).

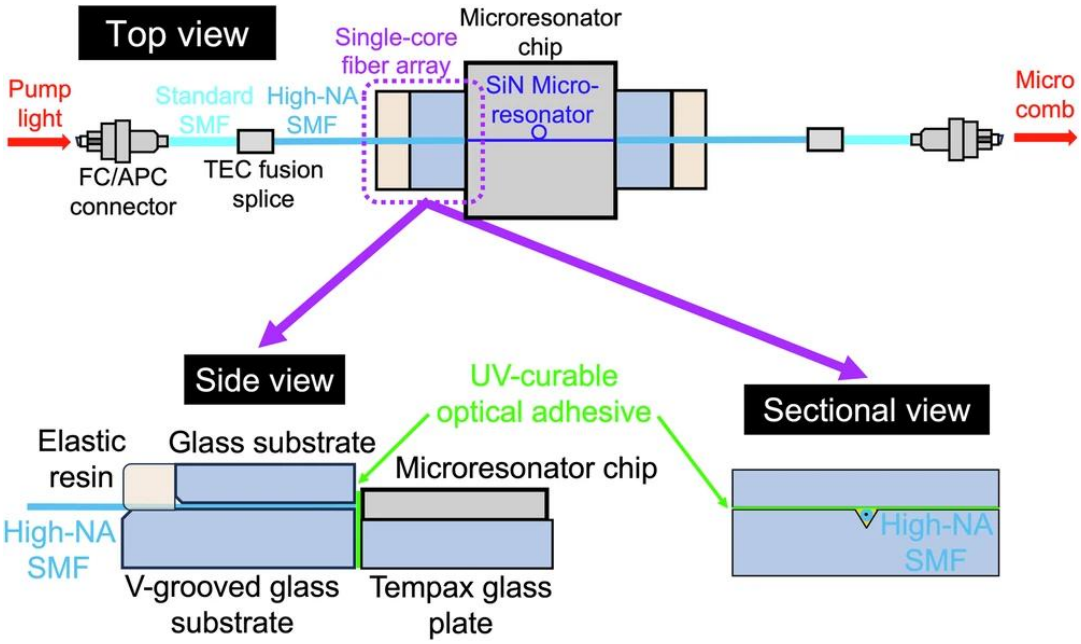


Figure 1. *The drive/clock chain, reproduced from Tokizane et al. [5] (CC BY 4.0): a fiber-coupled SiN soliton microcomb feeds bandpass filters selecting two comb lines; these injection-lock two DFB lasers (one IQ-modulated); the recombined beams photomix in a UTC-PD to emit the carrier. As demonstrated, the chain emits a CW (data-modulated) carrier; the pulsed clock of this design requires the multi-line extension described in Section 3.2.*

3.3 Launch, read-out, and in-situ reference

On-chip terahertz launch and detection follow the monolithic optoelectronic circuit of Kusyak et al. [7]: a generator photoconductive switch at the center of a coplanar stripline with two symmetric detector switches, providing odd-mode propagation, galvanic isolation between generation and detection, and — critically for a fabric that must be trimmed against fabrication spread — simultaneous in-situ measurement of signal and reference under identical conditions (Figure 2). The demonstrated switches are amorphous-silicon on sapphire, driven by 310-fs, 515-nm pulses; the demonstrated −3 dB bandwidth is 440 GHz, and its extension toward 1-THz carriers is an engineering assumption the bench protocol tests directly.

The fabric uses this one-launcher / two-detector / one-reference topology at the fabric edges for launch and Sum/Carry read-out, and reuses the detector/reference elements mid-fabric as the optical taps of the re-sync stations (Section 6.2). What no cited work supplies is the transducer from the stripline's micron-scale mode into the ~100×-confined acoustic-plasmon mode: the mode-area, impedance, and velocity mismatches make this the least-quantified link of the chain. Yoshioka et al. [4] demonstrate that photoconductive launch into gated graphene plasmons works (at 4 K); the efficiency at 300 K on this stack is assigned to the Boltzmann–Maxwell tier — the full kinetic-and-electromagnetic modeling level of Section 10, one rung above this paper's reduced-order chain — and to the bench (Sections 9, 10). A graphene patch antenna of the kind designed by Haque et al. [8] remains the free-space port option — noting it is a simulation-only design at 3.75–6.46 THz, cited as a style, not a demonstrated port.

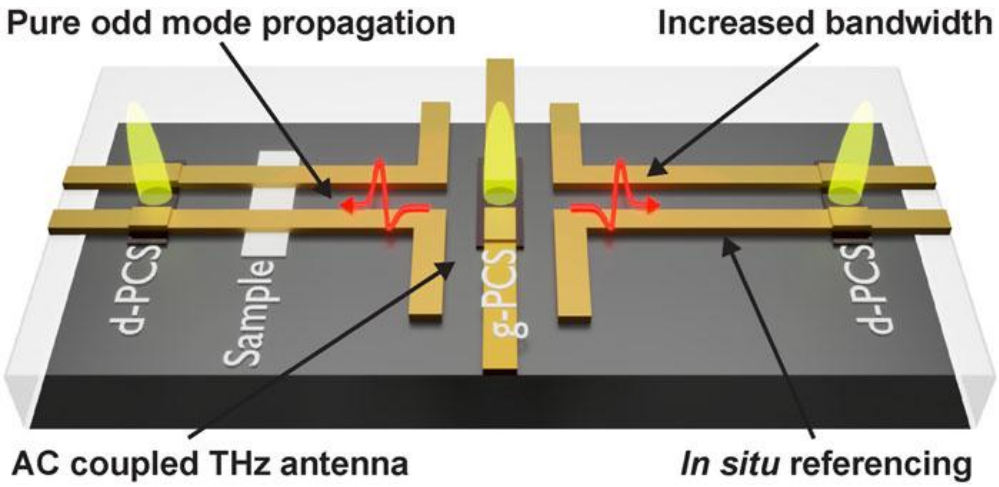


Figure 2. *Monolithic launcher/detector/reference circuit, reproduced from Kusyak et al. [7] (CC BY-NC 4.0): APL Photonics 10, 076117 (2025): generator switch (g-PCS) centered on a coplanar stripline with two symmetric detector switches (d-PCS) — galvanic isolation plus in-situ referencing. Used at the fabric edges for launch/read-out and reused mid-fabric as re-sync clock taps.*

4. Operating point

Parameter	Value	Note
Carrier frequency	~1 THz ($\hbar\omega \approx 4.14$ meV)	systems compromise: lithography and comb rate from below, drive/read-out chain from above (Section 1)
Symbol/clock rate	0.2–0.5 THz (0.25 THz baseline)	few-cycle pulses; set by comb line spacing. Clock-chain design spec — fan-out-2 logic closes in-model at ≈0.07 THz, 69 GHz (band 63–74 GHz), with a slot-confined, data-gated launch; neither confirmed nor moved at the ratio bias of Section 7.3 (Section 6.2)
Plasmon mode	acoustic / gate-screened (metal top gate)	required for confinement and gate programmability [4]
Quality factor $Q = \omega\tau$	6.3 (300 K) / 5.3 (80 °C)	$\tau \approx 1.0$ ps → 0.85 ps at the cap; best-case material (see text)
Carrier density n	10^{12} cm^{-2} (E_F ≈ 117 meV)	selected by disorder from below (on the design-model sharpness, Section 7.4), gain and heat from above (Section 7)
Plasmon speed s	2.2–2.3×10^6 m/s (design value 2.33×10^6)	Eq. (1) at d = 10 nm, ε_z(hBN) ≈ 3.4–3.8
Plasmon wavelength λ_p	≈ 2.3 µm	s/f_0
Logic cell	DS cavity, $L = (s/4f_0)(1-M^2) \approx 576$ nm	drift-corrected quarter wave, Eq. (4); M is the drift Mach number v_0/s (Section 5)

Drift bias	M/M_th = 0.7 (tracked ratio); M ≈ 0.10, $v_0 \approx 2.4\times10^5$ m/s at 80 °C	M_th ≈ 0.147 (80 °C), 0.125 (300 K), Eq. (3), on the Galilean drift law (Section 5); calibrated in situ by approach to self-oscillation, which presupposes a threshold — open at 353 K at the kinetic tier (Section 10)
Lithography class	~500 nm	deep-UV
Chip temperature	≤ 80 °C, no cryostat	locked requirement; 55 K rise budget over 25 °C ambient

Table 3. *The operating point. Temperatures are tagged throughout the manuscript: τ, Q, M_th and the loss figures all differ between 300 K and the 353 K cap, and every number is quoted with its temperature. The bias row carries the deepest commitment: the operating point is a tracked ratio to the measured threshold, not an absolute Mach number (Section 5).*

Three rows of Table 3 carry the design. The frequency row is the Q = ωτ argument of Section 1: at 1 THz the bit is a genuine wave, and passive loss (~4.3 dB per λ_p at 300 K, 5.1 dB at 80 °C; equivalently 27.3/Q dB) is steep enough to force distributed gain but shallow enough for a regenerative cell to beat.

The lifetime row is a best-case assumption, stated as such: τ = 1 ps at $n = 10^{12}$ cm^{-2} corresponds to mobility ≈ 86,000 cm^2/Vs at 300 K — inside the demonstrated envelope of hBN-encapsulated graphene (~140,000 cm^2/Vs near the acoustic-phonon limit [33]) but premium-material territory. The measured room-temperature baseline for propagating acoustic THz plasmons is shorter (τ ≈ 0.4–0.6 ps, Q ≈ 2.5–3.8 [39, 40]); the fundamental, phonon-set ceiling on plasmon lifetime in this stack is mapped in [42]; and gate-metal ohmic loss, not included in τ, adds to the budget at small d [41]. The row is also close to what the design needs. In-model, at the tracked bias the dissipation of Section 8.1 scales as τ^{-3} (Eq. (3) gives M_th ∝ 1/τ, and the Joule density is $n\cdot m^*\cdot v_0^2/\tau$), so, scaling the released ΔT ≈ 42 K of Section 8.2 (one-third fill, τ = 1 ps at 300 K, the assumed R_th), the 55 K budget closes in continuous drive only for τ(300 K) ≳ 0.92 ps (≈ 0.99 ps if the bias tracks the kinetic threshold of Section 7.5's closure), and at the measured 0.4–0.6 ps baseline only at ≈ 8–28 % duty (this paper's arithmetic on the released thermal model). That is a necessary condition, not a sufficient one: in the single-rate kinetic operator of Section 10 (A = 1, drift heat shed), a 353 K threshold is restored by a lifetime raised about ×2.6 but not by a d = 5 nm spacer or a ×6 higher f_0, while two of three admissible two-rate continuations of that operator restore one with no change of τ (in-model; Section 7.5). Where scalable (CVD-grown, centimetre- to wafer-scale) encapsulated stacks sit against this row in τ and Q at 1 THz, and the lowest Q at which the cell still closes its cascade budget, are open; the comparison is tracked as issue #140. The protocol's calibration sequence measures τ on-die, second only to the disorder σ (Section 10).

The density row is the most constrained choice, squeezed from both sides. From above: at $n = 10^{13}$ cm^{-2} the drift-gain route fails twice over — the saturation velocity caps the drift Mach number far below threshold, and the drive heat would allow < 1 % duty under the 80 °C cap (Sections 7.1–7.2). From below: at $n = 10^{11}$ cm^{-2} the logic swing becomes comparable to the charge-puddle inhomogeneity of even the best material, and cell yield

collapses on the design model's sharpness premise (Sections 7.3–7.4). n = 10^{12} cm^{-2} survives all constraints simultaneously.

5. One cell, all gates: the regenerative threshold logic family

Every logic gate in the fabric is the same object: a resonant DS cell — a gated-graphene cavity with asymmetric boundary conditions (source side density-clamped, drain side current-clamped [11, 19]) carrying a DC drift of velocity $v_0 = M \cdot s$, where the Mach number M measures the drift in units of the plasmon speed s — biased below its self-oscillation threshold (Figure 3a). This section states the cell's working relations as numbered equations; everything quantitative in the manuscript leans on them.

The acoustic-plasmon speed under a metal gate at distance d follows the gated local-RPA dispersion [39, 40]:

$$s = v_F (1 + g) / \sqrt{(1 + 2g)}, \quad g = 2e^2 k_F d / (\pi \varepsilon_0 \varepsilon_z \hbar v_F) \qquad (1)$$

with $k_F = \sqrt{(\pi n)}$ and ε_z the out-of-plane permittivity of the hBN spacer. At n = 10^{12} cm^{-2}, d = 10 nm and $\varepsilon_z \approx 3.4–3.8$, Eq. (1) gives $s \approx 2.2–2.3 \times 10^6$ m/s; the design value 2.33×10^6 m/s sits at the top of that band, and a ~5 % softer s would shorten L and the footprint proportionally without touching the threshold (see Eq. (3)). The constraint $s > v_F$ against Landau damping is met with factor ≈ 2.3 — and that margin rests on an identity, not an estimate: intraband particle-hole pairs on a Dirac cone satisfy $|\omega| \leq v_F q$ as a property of the spectrum, so single-particle Landau damping at the operating wavenumber is kinematically excluded for any electron distribution, drifted or not (demonstrated; Section 7.5). Two riders from the same tier attach to the equations of this section, both in-model and both quoted at the 80 °C cap. Eq. (1) is the collisionless speed: within the closure of Section 7.5 the mode runs ≈ 1.8 % slower at the operating point, so the fixed-length cavity needs a density trim — $\Delta n/n \approx +7$ % if that in-closure deficit is the whole of it — and a trim capability of that order is a design requirement rather than an option. The total speed shift is not settled by the closure, which does not compute its finite-temperature part; the kinetic operator Section 10 specifies, run in the bulk with a single relaxation rate, puts the bulk mode ≈ 11 % below Eq. (1) at 353 K, with a restoring trim near +48 % rather than +7 % (in-model; Section 7.5), so what the bench measures is open, and the trim capability must be sized against it (Sections 7.5, 10). And Eq. (3) below gives the hydrodynamic threshold, whereas the threshold the in-situ calibration measures is the physical one, if the cell has one. The closure of Section 7.5 puts it ≈ 12 % above the hydrodynamic value (in-model, within that closure). The same single-rate kinetic run finds none at 353 K when the electrons shed their drift heat: no crossing below the Landau ceiling — the drift at which the wave running against the drift slows to v_F — for $A \leq 1$ (in-model), where A multiplies the electron–electron rate estimate of [15] across an admissible band of 0.5–3; whether graphene's own electron fluid has a 353 K threshold is open (Section 7.5). The ≈ 12 % in-closure figure is a target for the next tier's solve rather than a bench expectation: a measured threshold also absorbs the contact non-ideality the band holds open (Sections 9.1, 10).

The DS small-signal increment for the asymmetric cavity — the per-second growth (positive) or decay (negative) rate of the cavity mode — with momentum relaxation included at the amplitude convention, is [11, 19]:

$$\omega'' = [(s^2 - v_0^2) / (2sL)] \cdot \ln[(s + v_0)/(s - v_0)] - 1/(2\tau) \qquad (2)$$

Setting $\omega'' = 0$ and expanding at small M gives the threshold in a form that exposes its invariances:

$$M_th \approx L/(2s\tau) \approx 1/(8\, f_0\, \tau) \qquad (3)$$

— independent of s and d once the cell is tuned to f_0, and proportional to temperature through $1/\tau \propto T$. At f_0 = 1 THz: M_th ≈ 0.147 at the 80 °C cap (τ = 0.85 ps) and ≈ 0.125 at 300 K. The drift also retunes the cavity: the fundamental of the biased asymmetric cell is

$$f_0 = (s/4L)(1 - M^2) \qquad (4)$$

so the 1-THz cell is slightly shorter than the zero-drift quarter wave (583 nm): L ≈ 576 nm at the operating bias.

Eqs. (2)–(4) carry the Galilean drift law: the two cavity legs travel at $\pm s + v_0$, and Eq. (2) composes them with momentum relaxation by the additive rule that [19]'s own onset condition also uses. Graphene's electron fluid does not follow that law. At small drift its drag is exactly ½ at every temperature in energy-conserving Dirac hydrodynamics — [19]'s own linearized speeds, $v_0/2 \pm v_p$, are this value at $T \to 0$ — and runs 0.35–0.69 in the kinetic and isothermal closures at finite temperature (both in-model). For a $T \to 0$ Dirac fluid with Eq. (1)'s speed pinned, the released friction law and the dissipated energy removed, the drift law alone moves the threshold ×1.67 above the exact zero of Eq. (2) (M = 0.2497 with the 353 K lifetime against 0.1493; ×1.74, M = 0.2201, at 300 K), still below the Landau ceiling (in-model; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md). Whether a threshold survives at finite temperature is the open question stated above (Section 7.5). The tracked-ratio specification below keeps its form; the drift law changes the bias current it implies (Sections 8.2, 9.2).

The operating point is a tracked ratio. Because M_th scales with temperature, an absolute Mach number is not a valid specification: a bias fixed at 0.7·M_th(353 K) crosses the ringing onset at 0.8·M_th(T) — the bias ceiling established below — after only ∼44 K of cooling. The specification is therefore the ratio M/M_th = 0.7, with M_th calibrated in situ by approach to self-oscillation (a calibration that presupposes a threshold; at 353 K its existence is open at the kinetic tier, Section 10) — at the 80 °C cap this corresponds to M ≈ 0.10, $v_0 = M \cdot s \approx 2.4 \times 10^5$ m/s. For the same reason, every solver operating-point result in Section 7 is reported relative to the solver's own threshold (Section 7.3; rows run at an absolute or analytic bias are labelled where they appear), never as an absolute Mach number (Figure 8(A) compares the two thresholds themselves on the analytic axis).

The cell's compact design model — the transfer the logic family is designed against, not a property the chain establishes for the released cell (see below) — is a sigmoid with saturating rails:

$$A_{out} = A_{pump} \cdot g(n_{local}), \quad g(n) = g_{max} \cdot \tfrac{1}{2}[1 + \tanh((n - n_{th})/\Delta n)] - \ell, \quad n_{local} = n_{bias} - \beta \cdot A_{in} \qquad (5)$$

where A_pump is the clock-pulse amplitude the cell amplifies, A_in the logic-input amplitude arriving from upstream gates, g_max the small-signal gain, ℓ the linear loss, n_th and Δn the threshold center and width, and β converts input amplitude into a local density shift. In the design model, logic follows from bias and coupling sign. The input couples to the cell's density through a gate segment, so its sign is a polarity choice: park n_bias above n_th with depleting coupling and a HIGH input pushes the cell below threshold (gain off → output LOW) — inversion as a bias choice. AND and OR are the same cell with enhancing coupling and the threshold applied to the sum of two input fields: a high threshold fires only on both inputs (AND), a low threshold on either (OR). Three quantitative properties make the family viable in this design model; how far the released chain supports each is stated with it, and what the released cell itself executes follows them:

Regeneration condition (design rule). The cell restores levels only if the threshold sharpness $k = \beta \cdot A_{swing}/\Delta n$ (A_swing is the rail-to-rail input swing) exceeds ∼5; the design rule is $k \geq 8$ with on/off extinction ≥ 10 dB, giving, by the released closed form (cell.static_noise_margin), a static noise margin of 0.256 of the logic swing at k = 8 and 10 dB extinction (0.338 at 20 dB; 0.221 under the fan-out-2 softening k_eff = 8/√2 of Section 6.3; 0.340–0.394 at k = 16 and 13–20 dB). The chain assumes this rule rather than deriving it from its cell — the closed form never calls the cell's transfer — and the released cell does not supply it (below).

Compression and re-clamping. The nonlinear solve shows 1-dB gain compression at a source/drive-plane amplitude $\delta\rho/\rho_0 \approx 1$ % (measured knee 0.76–1.15 % across the CW and pulse observables at N = 240, moving only −8 % (pulse) / −3 % (CW) by N = 480 on the ratio bias; saturation parameter A_SAT_eff ∈ [0.015, 0.023]) and, at a 10 % source drive, −9.1 to −12.1 dB of gain compression at N = 240–720 under the ratio bias this section prescribes (−9.1 to −13.7 dB under the absolute bias, a fixed M at every grid; demonstrated). The compression form extrapolates ∼14 dB there, which on the ratio bias neither the grids nor their continuum (−10.2 / −12.4 dB, pulse / CW, in-model) reach; only the absolute bias, whose fixed M sits at 0.77 of the grid's own falling threshold by N = 960, reaches it on the CW observable at N ≥ 960 (notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). It is a compression point, not an output rail: the released cell has no output plateau below near-depletion (demonstrated for the released Galilean solver at drives up to 0.7; above a ≈ 0.03 drive the model's fluid speed exceeds v_F, so this is a statement about the model, not the physical cell). The logic swing belongs at the knee, and what re-clamps the '1' level gate after gate is the cell's soft compression: a loaded fan-out-2 cascade settles near a 2 % drive (4–5 % at fan-out 1; in-model, read off the single-cell sweep under the cascade assumptions of notes/2026-08-01-gated-requirement-round-trip.md — no multi-cell cascade has been run). The cascade budget of Section 7.4 is the compression form evaluated at a knee-scale swing, not a cascade solve.

Bias ceiling. Above ∼0.8·M_th the resonator rings between clock pulses (5 dB pulse-to-pulse spread at 0.94·M_th): the usable window is 0.5–0.8·M_th — at the 80 °C cap, M ≈ 0.073–0.117 — with 0.7·M_th the working point (in-model, released solver; Section 7.3).

What the released cell executes. The design model's coupling constants (β, Δn, n_th and the pump) are not computed by the released chain; when executed, the released Galilean cell departs from Eq. (5) in its shape, in the sign and sharpness of its input coupling, and in having no two-field threshold (in-model on the released cell unless marked; notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md). Its own g(n) is a resonance curve with no plateau on either side, not a sigmoid: with its off state pinned at the 10-dB depletion (−0.20·n), its 10–90 % width is 0.15–0.20 of the density at the pump (Δn = 0.07–0.09·n on Eq. (5)'s tanh scale), with n_th 4–7 % below the operating density. Its only density coupling for a carrier-frequency input is rectification, and it accumulates rather than depletes — a delivered '1' raises the drain density by +0.86–1.06 of its swing with the cavity retuned and the bias held absolute (0.69–0.76 in the shipped tuning and at the analytic bias, 0.7 of Eq. (3)'s threshold) — so no logic input depletes the cell, and that coupling's sharpness is k = 0.18–0.26 on the tanh scale, far below the regeneration floor of ∼5. Inversion as a bias choice therefore holds only for an external bias swing (−0.20·n gives ≥ 10 dB on the drain density at fixed current, and −0.17 … −0.28·n is needed across the configurations where 10 dB is reached at all; on the absolute bias, which sits at 0.80 of the retuned continuum threshold, −0.20·n also brings the cell to its own threshold, which at G1's bias of 0.7 it reaches at −0.30·n), supplied by an electrical gate stage — ×8–17 gain behind a biased detector segment, ×37–73 behind an unbiased one, across the depletion band where 10 dB is reached (Section 9.8) — or by the optical tap of Section 9.6, not by a logic input. Under amplitude encoding the only inverter a logic input drives is destructive interference with a coherent clock pulse (demonstrated; retuned cavity, absolute bias), which reaches 10 dB only within 10.1–12.3° of phase error (by grid, tightening under refinement) and ±17.6–21 % of amplitude, with an operational (10–90 %) sharpness k ≈ 1.3 — 2.9 on the tanh scale on which the floor of ∼5 is read — so it fails the regeneration condition above. And the two-field threshold is absent: two summed '1's read only 2.6–3.3 dB above one (demonstrated; retuned cavity, absolute bias; those rows run where the model's fluid speed exceeds v_F, so they are the released model's reading) — the summed field acts as an OR with no margin, not as an AND; the cell compresses, it does not threshold.

Under a phase encoding — a 0/π bit on a carrier referenced to absolute time — NOT is instead a half-period (0.5-ps) delay, ≈ 2.6 dB of passive line at 353 K, and a $2f_0$ pump makes the cell a phase-sensitive amplifier that restores phase in a narrow window (knee-level swings under a 5 % pump; only weakly at the delivered '1'; in-model, notes/2026-10-17-a-pumped-cell-restores-phase-in-a-narrow-window.md). Its majority gate reads the sign of the summed field and needs no threshold, but keeps the model's fluid speed below v_F only for inputs of a third of the knee or less, about 3.8 quanta per unit, far from decision grade (in-model), and AND and OR then need a coherent f_0 reference, which was not run.

The released code itself carries Eq. (5) with A_in in place of A_pump as the prefactor (cell.py), a different map: composed with itself it supplies a family of attracting two-level states indexed by constants the chain never fixes, it has no stable empty slot, and no released result calls it (demonstrated, source read; notes/2026-08-02-zero-floor-spec-dissolved.md).

A precedent note, stated plainly: sub-threshold regenerative amplification of injected pulses by a DS cavity has no direct experimental or theoretical literature precedent. Its nearest relatives are regenerative resonance narrowing below threshold (demonstrated for detection [23]), amplified boundary reflection (standard DS theory [20]), and giant-gain grating-gate simulations [21]. The concept lives or dies at bench gate G1 (Section 10).

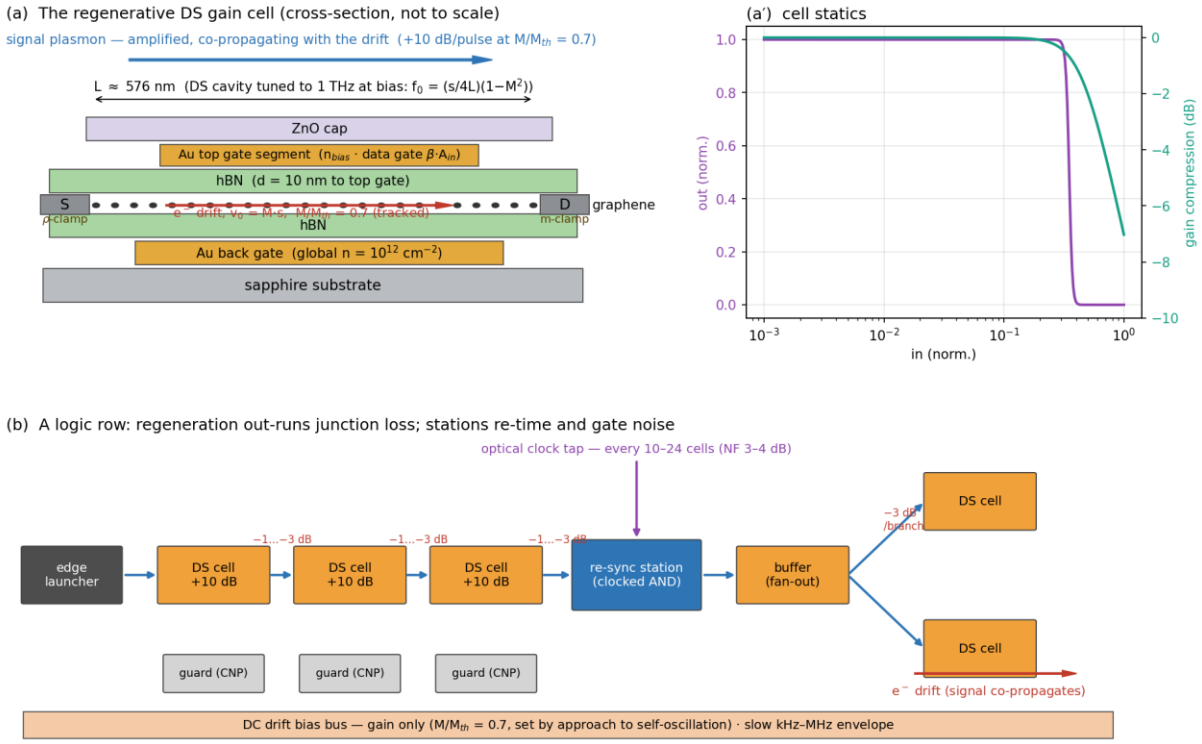


Figure 3. *The regenerative DS gain cell and its use in a logic row. (a) Cross-section on the Section 3.1 stack: the L ≈ 576 nm cavity between a density-clamped source contact and a current-clamped drain; the signal co-propagates with the electron drift (Section 6.1). (a′) Cell statics in the design model: the inverting tanh transfer of Eq. (5) and its saturating rail (the released cell's own transfer is a resonance curve with no plateau, Section 5). (b) A logic row: per-cell regeneration out-runs junction loss; a clocked re-sync station re-times pulses and blocks amplifier noise every 10–13 cells under the 20-dB link budget (Section 6.2); fan-out branches from buffer cells; CNP guard cells absorb strays.*

6. Architecture

Figure 4 assembles the whole machine: the photonic chain of Section 3.2 on the left generates clock and data pulses; the fabric of DS cells amplifies and routes them under the gate-map program; the DC bias bus below supplies gain only; the read-out elements of Section 3.3 sit on the right; and an electronic layer wraps the fabric for storage and trim.

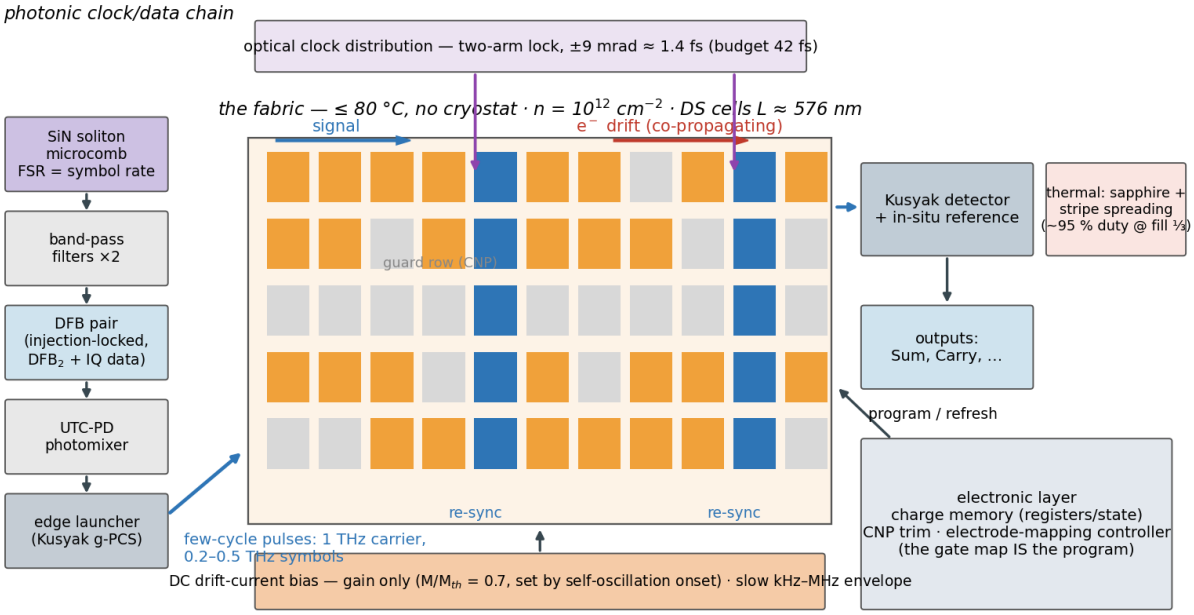


Figure 4. *System view. Left: microcomb → injection-locked DFB pair (IQ data) → UTC-PD photomixer → edge launcher; few-cycle pulses at 1 THz carrier, 0.2–0.5 THz symbol rate (proposed extension of the demonstrated CW chain, Section 3.2). Center: the ≤ 80 °C fabric — rows of regenerative DS cells (one cell = one gate) with re-sync stations fed by optical clock taps, CNP guard rows, signal co-propagating with the electron drift. Bottom: the DC drift bias (gain only, M/M_th = 0.7) with slow envelope gating. Right: monolithic read-out and the electronic charge-memory/trim layer.*

6.1 Gain and routing: one DC drift bias, two directional effects

A uniform DC drift bias supplies the gain; the gate map programs which cells couple to it. Cells parked at the charge-neutrality point are dark and absorbing — the gate pattern defines both the signal paths and the gain, suppressing cross-talk.

Two directional effects must be kept distinct, because they invite opposite intuitions. First, the DS gain is a cavity effect, not a traveling-wave gain: energy enters at the current-clamped drain boundary, whose reflection coefficient $(s + v_0)/(s - v_0)$ exceeds unity, and the cell amplifies its resonant mode as a whole, returning energy to both ports. The cell's directionality is a bias-orientation property — reversing the drift relative to the asymmetric boundaries turns the increment of Eq. (2) into a decrement — not a property of the signal's transit direction. Second, the drift partitions the bulk damping nonreciprocally: both branches of $\omega = (s \pm v_0)k$ decay at the same temporal rate $1/(2\tau)$, so the spatial attenuation is

$$\kappa\pm = 1 / [2\tau(s \pm v_0)] \qquad (6)$$

per unit length — the wave co-propagating with the electron drift is the less damped one, in agreement with the drift-nonreciprocity literature and the Fizeau-drag experiments [24–27]. Signal routing therefore follows the electron drift. At the operating bias the contrast $(s + v_0)/(s - v_0) \approx 1.2–1.3$ amounts, via Eq. (6), to ≈ 0.26 dB per cell transit at 300 K (≈ 0.31 dB at 80 °C), on the Galilean drift law of Eqs. (2)–(4), which graphene's electron fluid does not

follow (Section 5) — a useful forward bias, but an order of magnitude short of isolation, and this design claims no built-in isolator from the drift.

Suppression of reflective loops is accordingly not claimed from the drift. It rests on four architectural mechanisms: the pulsed clock (gain exists only during clock slots), the noise-flushing re-sync stations (Section 6.2), the absorbing CNP guard cells, and junction loss. The residual round-trip stability of coupled cells through the charge channel is checked in Section 7.5 under a closed-form coupling ansatz; the radiative coupling channel remains with the Boltzmann–Maxwell tier (Section 10).

6.2 Clock: the launcher pulse train

Timing is generated at the fabric's left-edge pulse generator — the Section 3.2 chain — emitting few-cycle 1-THz pulses at a 0.2–0.5 THz repetition rate. (At the fan-out-2 operating point of Section 6.2(iii), transient gain is set by drive duration, not slot length, so the launch burst must last most of its slot — and must also end before the slot does: a drive that fills its slot hands the next slot an un-decayed field, and a '0'-slot floor ≥ 10 dB below the data forces a flat, data-gated burst of duty ≤ ≈ 0.8, with raised-cosine edges of two round trips and the '0' slots undriven (in-model; notes/2026-07-31-physical-launch-gated-frontier.md). That launch costs ×3.4–4.9 the energy per '1' of the released 3-cycle burst but only ×1.25–1.30 in time-averaged launch power (demonstrated waveform integrals); the cost falls on the launch subsystem, outside the fabric-side energy of Section 8.1.) The DC bias carries no timing; it is gated only by a slow envelope (kHz–MHz, comfortably inside the bias-plane RC corner) for power management and noise flushing. The clock source's phase budget closes with ~30-fold margin; the cells' own transfer phase draws on the same ±15°, which makes trim and bias tracking phase requirements as well (Section 3.2).

Three riders attach. (i) Re-sync stations: a launcher-only clock is a source-synchronous wave pipeline, so jitter accumulates; clocked-AND re-timing cells, fed optically by the same comb through mid-fabric taps (Section 3.3), are inserted every 10–13 cells — the spacing set by the noise budget: with a launch SNR of 20 dB, which the released law reads as a link budget over the thermal floor of Section 7.4, and 6 dB margin, the accumulated noise (≈ NF + $10 \cdot \log_{10}$ N over N restoring cells) holds station spacing to ~12/6/3 cells for per-cell noise figures of 3/6/10 dB — 13.3 cells at the 2.77-dB floor and 10.0 at NF = 4 dB; no documented margin convention reaches 24 cells, which would need NF ≈ 0.2 dB, below the floor (notes/2026-08-25-what-the-20db-launch-budget-denominates.md). NF ≲ 4 dB is the requirement this law sets on the cell (Section 7.2); within Eq. (7) the bare cell meets it at any gain, so it binds on what precedes the gain and on the amplifier's idler (Section 7.4). What the 20 dB denominates is itself open: if the coded launch jitter of Part II [46] (10 % of the full swing) is physical at its coded reference plane, the same law gives 4.4–7.8 cells for NF 2.8–4 dB at 300 K, and across reference-plane readings the spacing at NF 2.8 dB runs from about one cell to 12.2, a ×12 spread (in-model; Section 9.5). (ii) ASE and ring-down flushing: the clocked re-sync stations pass energy only during their clock slots, flushing both accumulated amplified-spontaneous-emission (ASE) noise and residual cavity field. The DC bias itself carries no per-slot timing — it is gated only by the slow (kHz–MHz) envelope above, far below the symbol rate — so this flushing is the stations' job, not the bias's. (iii) Inter-symbol interference (ISI): two timescales matter. Carrier-density recovery

(∼τ ≈ 0.85 ps against 2–5 ps slots) leaves pattern-dependent gain negligible on its own (< 0.1 dB spread at the 0.25 THz baseline in the design model's depleting transfer, Section 7.2). The binding effect is the regenerative cavity's ring-down: at the working bias it is ∼4–5 ps, comparable to the slot, so residual field carries pattern memory from one slot to the next. The same ring-down lifts the per-pulse gain toward its streaming value (Section 7.3) — and the two bookings are one effect: the accumulation that lifts the gain is the pattern memory. In-model the loaded ring-down ISI at the 4-ps slot is ≈3–4 dB peak-to-peak, and neither the cell's compression nor the stations can absorb it — gain compression at the operating drive is ≈0.06 dB, far too shallow to clamp a swing of that size, and the netlist's one re-sync station sits at the read-out boundary while containing per-slot memory takes a per-slot, per-cell flush. Per-slot flushing is necessary but not sufficient at this slot: no de-Q rate, including an instantaneous one, closes the fan-out-2 ledger at 4 ps, because the cavity's rebuild time binds. Under the pulse-knee convention, with a stretched launch burst, three configurations close near ≈0.1 THz: no flush at all at 79–92 GHz, a passive guard at 98–128 GHz, and an active de-Q at 100–148 GHz (worth ≈+10 % of clock over the passive guard, and 0 % at its 81 GHz demonstrated-calibrated corner). For a physical launch that convention is the wrong corner, because the compression knee migrates with drive duration. With the launch confined to its slot and gated by the data (the duty-0.8 burst of the clock rider), the no-flush configuration — which needs no new device element, only launch-waveform discipline — closes fan-out-2 logic at ≈0.07 THz: 69 GHz (band 63–74 GHz), and 67–71 GHz at the demonstrated grid rungs (worst-'1' gain 7.68 dB at 71 GHz and 7.93 dB at 67 GHz, N = 720, demonstrated; the frontier an in-model composition; notes/2026-07-31-physical-launch-gated-frontier.md), the 71-GHz edge holding only under a round ≥ 10-dB '0'-floor specification (the noise-margin-anchored 11.8-dB variant shrinks the window to 67 GHz alone). At 91 GHz a slot-filling burst breaks the 1-dB ISI criterion (1.24 dB) and lifts the '0'-slot floor to −5.2 dB (demonstrated), leaving that slot to fan-out-1 wires; the guard and de-Q configurations have not been re-adjudicated with a physical launch (open). These rows are biased at 0.7 of the analytic threshold. At the ratio to the solver's own threshold that Section 7.3 prescribes, the gain clears the measured requirement by +1.1 to +1.3 dB but the '0'-slot floor becomes the binding leg, and 69 GHz is neither confirmed nor moved; which bias convention the adjudication should use is open, because the two converge to different biases (0.7 × the inviscid against 0.7 × the analytic threshold), a permanent 1.5 % design fork that must be settled as a named design decision (notes/2026-08-01-ratio-bias-gain-table.md, notes/2026-08-01-gated-requirement-round-trip.md). The 0.2–0.5 THz range remains the clock-chain design spec; what moves is the slot at which fan-out-2 logic closes. The single quotable ISI figure is convention-dependent and remains open for the bench, and so does the '0'-slot floor on which a gated slot rate turns: the released solver reads that floor as an intracavity velocity maximum near the source, while the swing it is specified against is a drain-plane density (at 67 GHz the two readings differ by +0.055 dB at N = 240 and −0.445 dB at N = 480, demonstrated), and the released compact logic map supplies a family of extinctions rather than a bound, so the floor specification must be imported (notes/2026-08-02-zero-floor-spec-dissolved.md); the gated frontier is not moved by it (Section 10; community record: notes 2026-07-22 through 2026-08-02).

6.3 Fan-out

A Y-split costs 3 dB per branch plus junction loss. At the working point the regenerative cell delivers ~+8 dB per pulse (Section 7.3), so fan-out of 2 with restoration headroom closes: +8 dB against 3 dB split plus 1–3 dB junction loss leaves 2–4 dB of margin before the compression knee (the released cell has no output rail, Section 5). The margin is real but modest — and the kinetic correction of Section 7.5 spends part of what is left. Every gain figure in this paragraph is a hydrodynamic-tier figure, and the correction reads them all down: −0.95 dB per cell centrally at 353 K, band [−1.5, −0.3] dB, in-model within the closure of Section 7.5 (whether a 353 K tracked point exists at the kinetic tier is open there), with a separate open ±1 dB boundary allowance riding on top. Within the closure the direction is unambiguous and the conclusion survives — fan-out-2 still closes across the signed band, but on margin rather than on comfort, so margin management becomes load-bearing rather than optional: reserving the low-loss junction class for splits, and spending buffer tiles instead of headroom. No corrected fan-out margin figure is quoted here, and the omission is deliberate. Per-cell gain in this fabric is defined at the junction-attenuated swing (Section 7.4), so it rises with junction loss and cannot be netted against a junction budget by subtraction; a corrected margin has to be recomputed per junction class against the cascade budget rather than read off the band, and the tier that owns the correction (Section 10) is what will settle it. The promoted composed envelope supplies the 0-dB-junction in-model reference point — a fan-out-2 reserve of +2.01 dB [1.46, 2.66] near-specular CW (+2.61 pulse), effective sharpness k_eff = 8·F^(−1/2), and a disorder ceiling F ≤ 2.56 — computed without junction attenuation, so it complements rather than replaces the per-class recomputation. Its effective sharpness and disorder ceiling inherit the design $k_0 = 8$ of cell.py, which the released cell does not deliver (its only density coupling gives 0.18–0.26 on the same scale), so the disorder-floor ceiling applies to no released cell, while the +2.01 dB reserve rests on the gain ledger and is untouched (notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md). The reserve is also a CW quantity. At the gated slot rates of Section 6.2(iii) a different quantity has been measured: the existence headroom of a duty-0.8 launch — its delivered gain less the 6.515-dB small-signal fan-out-2 requirement, a criterion that reserves no margin — is +0.98 dB at 71 GHz and +1.21 dB at 67 GHz (N = 480; analytic bias, near-specular contact, bulk term central) and turns negative at 71 and 77 GHz at the −1.00 dB contact corner (demonstrated rows, in-model composition; notes/2026-08-01-gated-requirement-round-trip.md); being a gated-drive quantity it does not grade the CW reserve, and the bias convention behind it is open. At the 4-ps slot the CW reserve's closure is additionally conditioned on the flush question of Section 6.2(iii). Higher fan-out uses dedicated buffer tiles in any case (Figure 3b).

6.4 Memory: compute in plasmons, store as charge

Stated plainly: there is no native plasmonic storage at ~1 ps lifetimes. The bit's persistent home is the charge on a cell's gate (DRAM-like, slow, refresh-limited); pumped bistable cells and recirculating loops could serve only small fast registers; no register built from either is modeled or priced in the released chains. Driven with an added $2f_0$ pump at one clamp, the released solver's cell under phase encoding is a pumped bistable cell of this kind (in-model, 353 K): above an onset in drain-current modulation of (0.70, 0.72] of the DC drain current on N = 240, rising with refinement to (0.72, 0.80] on N = 720, it is a latching parametric oscillator that holds a $0/\pi$ bit under a continuous pump; neither the onset nor

the latched level converges on the grids run, whether a contact can impose that modulation at 2 THz is open, and gated in the 69-GHz slot of Section 6.2(iii) it amplifies a bit stream's flips ×1.75–3.15 over an unpumped cell but re-levels none (notes/2026-10-17-a-pumped-cell-restores-phase-in-a-narrow-window.md). No write–hold–read register cycle has been run, and Part III's lane uses none: it regenerates every waiting bit at every level it waits (Part III, Section 4.3 [45]). The fabric is a plasmonic datapath wrapped in an electronic memory layer — a hybrid by design, not by concession. The interface devices of that layer (charge sensing, refresh, level conversion to the launcher drive) are conventional electronics and are itemized in Section 9. Beside the optical launch and read-out of Sections 3.3 and 9.5, whether a bit can cross directly between this layer and a plasmon slot is an open design question, tracked as issue #141. Two couplings for it are priced on the released cell: the cell's own rectification, by which a delivered '1' raises the drain-plane DC density by +1.7–2.1 % (demonstrated; retuned cavity, absolute bias), reaching 90 % in 6.9 ps and carrying 21–35 % of itself into the next 69-GHz slot (in-model); and a gate depletion of the clock-driven cell, which needs −0.17 … −0.28·n for 10 dB of extinction, in the configurations where 10 dB is reached at all (in-model). The gate's RC is not modeled (notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md).

6.5 The machine: a half adder

The half-adder block computes the textbook decomposition S = (A OR B) AND NOT(A AND B), C = A AND B, mapped onto five active cells — OR (low threshold), AND (high threshold, doubling as the carry source), NOT (inverting bias), one delay buffer, and the output AND — plus two optional buffers that align the carry to the sum's slot (Figure 5). Both fan-outs in the netlist are fan-out-2, exactly the budget Section 6.3 provides. The two inputs of the output AND arrive through equal depth (OR→DLY vs AND→NOT), so the design needs no asymmetric timing trim; total logic depth is three cells. Operation is wave-pipelined: successive additions follow one another through the fabric one clock slot apart, with no latches between gates. Table 4 gives the truth table including the intermediate signals; Table 5 collects the block specification. The gate kinds are those of the design model of Section 5: in the released cell the inverting-bias NOT has no density mechanism at the delivered swing (in-model), and the summed-field threshold separates '1 + 1' from '1' by only 2.6–3.3 dB (demonstrated; retuned cavity, absolute bias), so Table 4 is the netlist's Boolean logic rather than a released-cell result (Section 5). Latency is likewise a design value. In-model the released cell delays a logic burst by picoseconds per cell: on the 3-cycle clock pulses of the 0.25 THz design baseline (Section 7.3), 1.1 ps per cell by the half-maximum rise and 3.2–3.4 ps by the energy centroid (measured by the note's reviewers on that waveform and recorded in notes/2026-10-13-phase-is-not-the-easier-half-for-the-cell.md), 2.0–4.4 ps for a 12-ps gated burst and 4.4–5.9 ps for a narrowband envelope — so depth 3 takes ≈3–10, 6–13 and 13–18 ps respectively.

A	B	OR	AND (= C)	NOT	S
0	0	0	0	1	0
1	0	1	0	1	1
0	1	1	0	1	1

1	1	1	1	0	0

Table 4. *Half-adder truth table. The OR, AND and NOT columns are the intermediate signals at the three internal cells, as the design model of Section 5 specifies them; S and C are the read-out ports.*

Metric	Value	Derived in
Throughput	one addition per 4-ps slot = 2.5×10^{11} additions/s (wave-pipelined; 0.25 THz design baseline — fan-out-2 logic closes in-model at ≈0.07 THz, 69 GHz (band 63–74 GHz), with a slot-confined, data-gated launch; neither confirmed nor moved at the ratio bias of Section 7.3, Section 6.2)	Section 6.5
Latency	design value ≈ 1 ps (depth 3, sub-slot); in-model on the released cell ≈ 3–10 ps at depth 3 for the 3-cycle burst, 6–13 ps for a 12-ps gated burst, 13–18 ps narrowband	Figure 5c; Section 6.5
Logic depth / fan-out	3 cells / ≤ 2 everywhere	netlist, Figure 5a
Worst-path gain budget	launch → −3 dB split → +8 dB cell → −1…−3 dB junction → +8 dB cell → junction → +8 dB cell → read-out; each stage pulls the '1' level toward the loaded fan-out-2 fixed point near a 2 % drive, contracting its distance from that point by a factor 0.51–0.59 per gate (the cascade map's local slope; in-model; no multi-cell cascade run) — compression, not a rail: at a 10 % drive the cell gains only 1.3–1.6 dB (demonstrated) against the 6.5 dB a fan-out-2 stage needs	Sections 5, 6.3, 7.4
Footprint	~11 × 7 lattice sites ≈ 12 × 16 μm incl. guards, clock tap, read-out boundary	Figure 5b
Drive power (7 biased cells: 5 logic + 2 carry-alignment buffers)	~0.09 mW at the operating bias (1.2 kW/cm^2 × 7 × 10^{-8} cm^2); up to ~0.12 mW at the upper calibration bound	Section 8.1
Energy per addition (fabric side)	≈ 0.35–0.5 fJ at the 4-ps baseline slot (scales with slot length; Section 8.1)	Section 8.1
Internal re-sync stations	none needed (depth 3 < the 10–13-cell spacing of the 20-dB link budget, Section 6.2); one station at the read-out boundary	Section 6.2

Table 5. *Half-adder block specification. The drive-power and energy ranges span the analytic operating bias and the upper, solver-class calibration bound; Section 8.1 derives both.*

The block's defining property is that its outputs are re-timed at the boundary station and can drive the inputs of a following block. Their level depends on the regime, which is open: in the clocked regime of Eq. (5) they leave at the launcher's level; without a re-launch, a loaded fan-out-2 cascade pulls the '1' toward its fixed point near a 2 % drive (in-model, on the cascade assumptions of Section 5; no multi-cell cascade has been run); neither is a rail-

scale swing, which no run of the released chains or the promoted notes delivers to a deciding input (notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). Two half adders and an OR compose a full adder, and full adders compose a ripple-carry word. Composition on this latch-free fabric is not free, because every waiting wire must be regenerated at every level it waits: in Part III's 32-wide block-sum lane, built from these full adders, that costs 2,616 balancing buffers (+51 % on 5,160 logic cells at zero skew tolerance) and 393–679 re-sync stations, 8,169–8,455 cells in all (in-model; Part III, Sections 4.2–4.3 [45]). Nothing about the five cells is special to addition: they are the generic gate vocabulary of the fabric's design model (threshold-AND/OR, bias-inversion, buffering; Section 5 records what the released cell supplies of it). The half adder is simply the smallest machine that exercises all of it — which is why it, and not a larger demonstrator, is what the feasibility chain of Section 7 and the bench protocol of Section 10 are sized against.

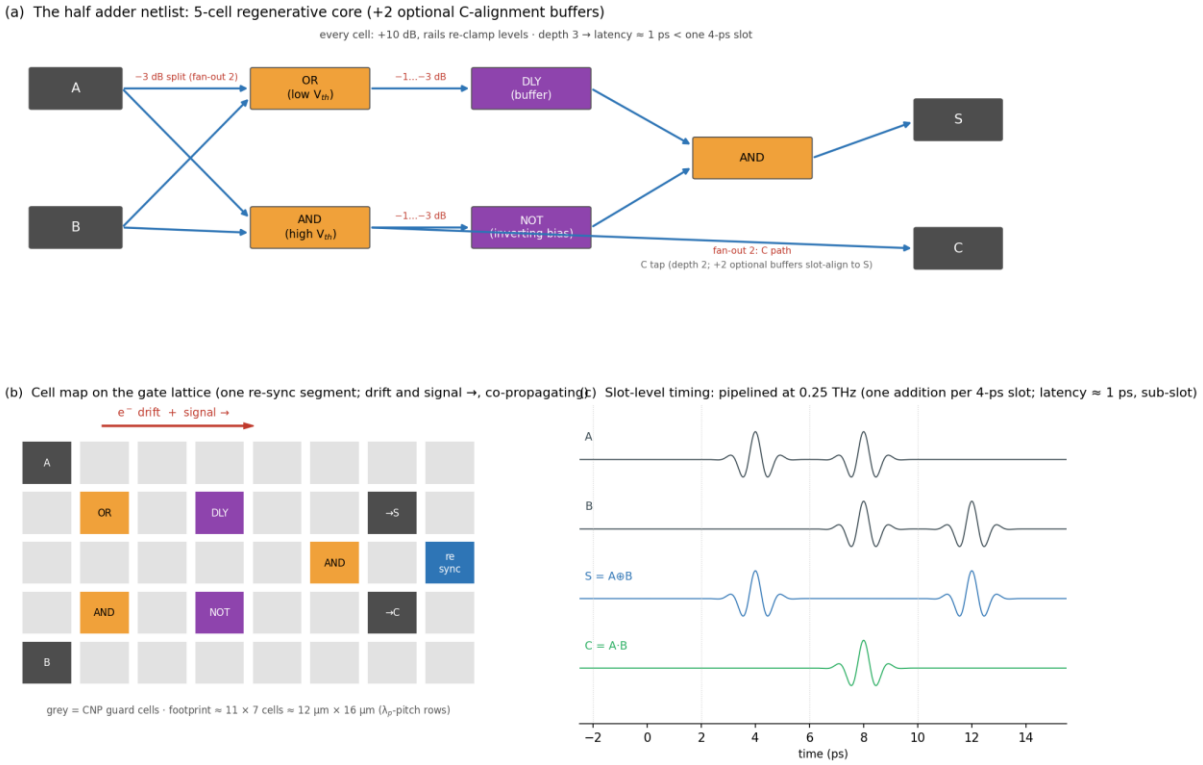


Figure 5. *The half-adder block. (a) The netlist: five regenerative cells (OR, carry-AND, inverting-bias NOT, delay buffer, output AND — the design model's gate kinds, Section 5) with fan-out-2 splits and per-edge junction losses annotated; two optional buffers slot-align the carry. (b) The cell map on the gate lattice — one re-sync segment; signal and electron drift co-propagate; gray cells are CNP guards. (c) Slot-level timing at the 0.25 THz design baseline (with a physical, data-gated launch, fan-out-2 logic closes in-model at ≈0.07 THz, Section 6.2): a pipelined add stream with S = A XOR B and C = A·B, drawn at the design latency; in-model the released cell's delay is ≈1–6 ps per cell by estimator and burst (Section 6.5).*

7. Feasibility: the five-pass chain

Each pass is a deliberately transparent reduced-order model. "In-model" throughout this manuscript means: established within this chain, under its stated assumptions, and not yet measured. The chain is ordered so that each pass either kills the design or sharpens the next pass's question — in brief: Pass 1 rules out brute-force bulk gain and selects resonant

cells; Pass 2 finds the low-density window where a resonant cell breaks even; Pass 3 locates the regenerative operating point; Pass 4 budgets cascade, noise, and design-model yield; Pass 5 bounds the model's own validity. This section reports results only; the governing equations, conventions, and the runnable solver for each pass are collected in Appendix A. All bias-dependent results are reported relative to threshold (Section 7.3).

7.1 Pass 1 — bulk gain and brute thermal limits (the double no-go)

A drift-Doppler cascade model established two independent failures at the highest candidate density, n = 10^{13} cm^{-2} — the natural first choice for a robust, well-confined plasmon. Bulk convective gain is impossible: the density-dependent saturation velocity v_sat(n) ≈ ω_OP/k_F ≈ 4.5–5.4×10^5 m/s at 10^{13} cm^{-2} [9] (against s ≈ 4×10^6 m/s there), with ω_OP the graphene optical-phonon frequency, caps v_d/v_p at ≈ 0.13, so uniform drift can at best halve the per-gate loss — transparency is unreachable. Independently, the drive heat at that density (~190 kW/cm^2 at v_sat) caps the duty cycle at well under 1 % at any fill convention under the 80 °C budget. Both failures point the same way: lower density, resonant cells. The demonstrated amplification of [2] is consistent with this conclusion — it is a resonant, boundary-mediated effect by any of its candidate explanations (Section 2), not bulk traveling-wave gain.

Figure 6 — Pass 1: bulk gain and brute thermal limits (n = 10^{13} cm^{-2}, 353 K)

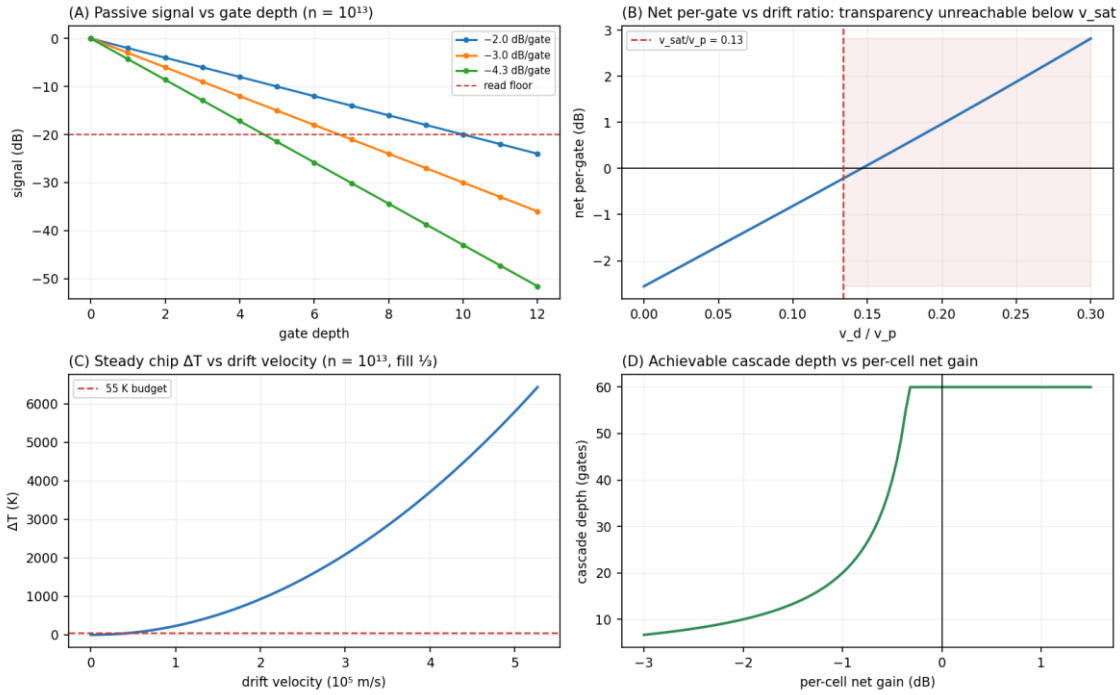


Figure 6. *Pass 1, at n = 10^{13} cm^{-2} and 353 K. (A) Passive signal vs gate depth for the candidate per-gate budgets. (B) Net per-gate result vs drift ratio v_d/v_p: transparency is unreachable below v_sat. (C) Steady-state chip ΔT vs drift velocity against the 55 K budget. (D) Achievable cascade depth vs per-cell net gain. Per-gate loss convention: one λ_p/2 of propagation (Appendix A).*

7.2 Pass 2 — analytic resonant cells: break-even only at low density

Applying the DS increment, Eq. (2), to a 1-THz-tuned cell shows that a single, non-regenerative resonant cell can at best reach break-even. Biased as hard as the saturation velocity allows, the amplified drain reflection $(s + v_0)/(s - v_0)$ only cancels the round-trip loss: net round-trip gain hovers near 0 dB across the whole low-density window and turns negative once v_sat caps the achievable drift below threshold — about −0.3 dB by 10^{13} cm^{-2}, where v_sat/s ≈ 0.13 holds M/M_th near 0.9. The low-density requirement is therefore a gain requirement, not only a thermal one; but reaching break-even is not reaching the +8 dB the logic needs, and a single transit never closes that gap. That is the work of the multi-pass regeneration of Pass 3. This pass also set the logic-margin design rule (k ≥ 8, extinction ≥ 10 dB — a rule of the design model that the released cell does not supply, Section 5), bounded the design model's carrier-recovery ISI as benign, and derived the re-sync spacing law quoted in Section 6.2, which sets NF ≲ 4 dB as the requirement on the cell — one the bare cell meets at any gain, so that it binds on what precedes the gain and on the idler (Section 7.4). Fan-out-grade single-transit gain (a 3 dB split, 1–3 dB junction loss, and restoration headroom) was unreachable without regeneration — a gap the next pass closes.

Figure 7 — Pass 2: analytic resonant cells (353 K)

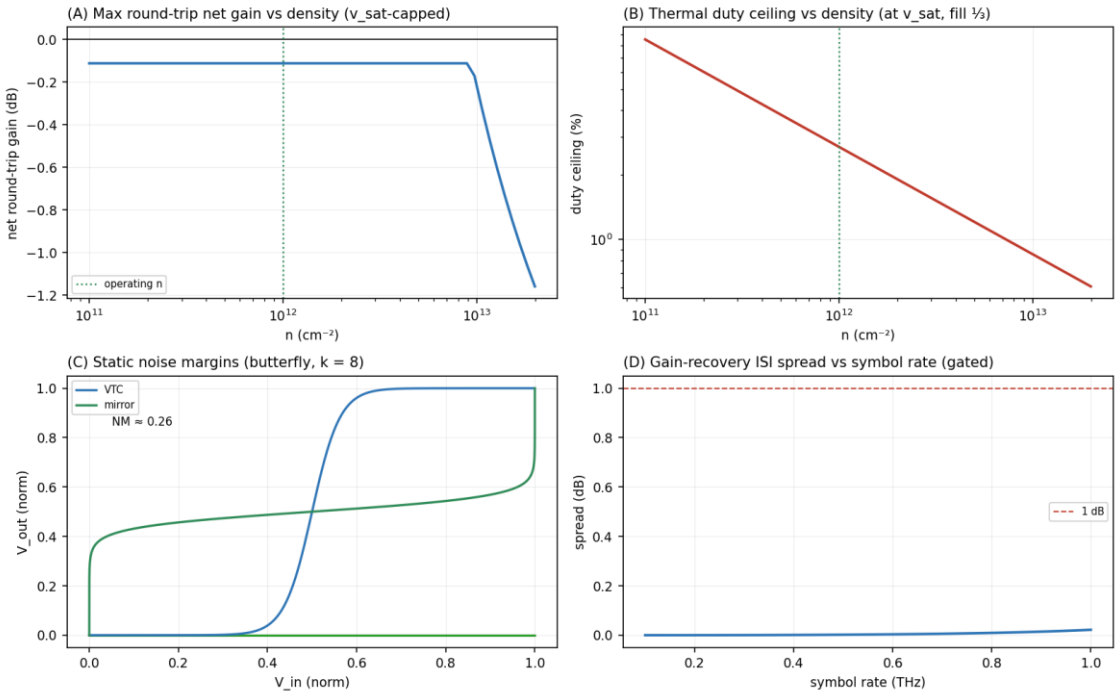


Figure 7. *Pass 2, at 353 K unless noted. (A) DS single-transit net gain vs density, Eq. (2). (B) Thermal duty ceiling vs density, refined from Pass 1. (C) Static noise margins (butterfly construction) of the design model's tanh cell at 6/10/20 dB extinction. (D) Pattern-dependent gain spread vs symbol rate at τ = 0.85 ps, in the design model's depleting transfer.*

7.3 Pass 3 — the regenerative operating point (nonlinear time-domain solve)

A nonlinear shallow-water solver with the DS boundary conditions confirms what the transit picture misses: below its oscillation threshold the cell is a multi-pass regenerative amplifier. The cavity transfer is a Fabry-Perot below threshold — the current-clamped

drain reflects with amplitude $(s + v_0)/(s - v_0) > 1$, the density-clamped source with $|r| = 1$, and the round-trip loop closes self-oscillation at loop gain unity, which recovers M_{th} of Eq. (3) exactly. The solver's measured threshold, $M_{th,num} \approx 0.169$, sits about 15 % above the small-M analytic 0.147 of Eq. (3). About a tenth of that offset is not numerics at all — it is the small-M expansion of Eq. (3) measured against Eq. (2)'s exact zero, the inviscid ≈0.1493 — and the rest is the released scheme's numerics, derived from first principles in its exact linearized operator (community notes, July–August 2026; shipped as runnable listings): 95 % bulk Lax–Friedrichs diffusion of density and momentum together (with an interior remainder whose mechanism is open), 4.8 % first-order placement of the two DS clamps and 0.15 % operator splitting, at N = 240 and cfl = 0.4 for the eight-seed, 240-round-trip median (in-model; notes/2026-08-07-mth-num-excess-decomposition.md). The excess converges at first order in Δx (fitted order 1.04–1.06 on eight-seed medians, demonstrated arithmetic on three ill-conditioned rungs; notes/2026-08-06-mth-num-record-length-and-fit-convention.md) toward the exact inviscid threshold ≈0.1493 — 1.5 % above the small-M analytic 0.147083, which the discrete operator does not recover ($M_{th,lin}$ = 0.1688/0.1588/0.1540 at N = 240/480/960); the corresponding released-chain rerun at N = 480 measured $M_{th,num} = 0.158748$, confirming the pre-registered 0.1590 ± 0.0010 (its eight-seed median is 0.15883), and the N = 720 rung, 0.155568, extrapolates with it to within 0.07 % of the inviscid limit, a pair-dependent agreement — the convergence claim is demonstrated at the released-chain level (Appendix A) — so the chain's own solver results are quoted at the ratio $M/M_{th,num}$ (the gated-slot rows of Sections 6.2–6.3 and 10, which the promoted notes ran at 0.7 of the analytic threshold, and the absolute- and analytic-bias rows of Sections 5–7 are labelled where they appear; which convention adjudicates the gated slot is open, Section 6.2(iii)) — a numerics-internal convention, not a physical bias fraction: with a physical bulk viscosity in place of the numerical one, the kinetic threshold within the closure of Section 7.5 sits at 0.165 [0.154, 0.169]. Threaded to finer grids, the released threshold scan stops: its floor, $1.05 \cdot M_{th}(353\ K) = 0.1544$, lies 3.43 % above the inviscid limit, so at N = 920 and 960 it returns no value at every seed tested while N = 860 stays finite (demonstrated; the failing grid lies in (860, 920], in-model; notes/2026-08-02-where-the-released-threshold-bracket-stops-working-a-scanned.md), and a released rung beyond that needs a widened scan floor first. Net gain referenced to the identical passive cell rises from break-even (cancelling the 2.56 dB per-gate loss — one $\lambda_p/2$ of propagation, Appendix A) at $M/M_{th} \approx 0.3$ through +9.7 dB continuous-wave at $M/M_{th} \approx 0.7$ toward formal divergence at threshold. The cell is low-Q ($Q \approx 6$), so an isolated few-cycle pulse does not fully charge it: the analytic recirculation of a single 3-cycle burst yields ≈ +6.3 dB, about 3.4 dB below the continuous-wave ceiling. Driven with the actual clock waveform — 3-cycle pulses at 0.25 THz, a ≈ 75 %-duty train in which each pulse rides the cavity field left by its predecessors — the solver delivers ≈ +8 dB per pulse in steady streaming operation. For slot-rate logic that figure decomposes: a data-independent worst-case-'1' gain of ≈+3.0 dB [2.9, 4.0] plus a ≈+4.8 dB predecessor pedestal contributed by the residual cavity field — the same accumulation booked as inter-symbol interference in Section 6.2(iii), so the two cannot both be banked. Community analysis (July 2026; the promoted calibration notes in the project repository) shows that for this streaming train the residual gap to the continuous-wave figure is not buildup physics ($\lesssim 0.1$ dB in an energy metric). The exact linearized operator of the released solver — constructed term-by-term from its update and ghost-cell algebra and shipped as a runnable listing in the

notes — settles the split: linear numerics (drive detuning from the zero-drift cavity length, interior Lax-Friedrichs diffusion, and a drift-induced boundary loss at the drain's discrete reflection) account for 0.956 dB of the 1.864 dB gap, and the remaining 0.908 dB is pulse-train structure, drive-amplitude compression near the 1 %-swing knee, and the per-slot estimator (a drive-amplitude sweep of the released solver since resolves that split as ≈0.87 dB pulse-train/per-slot-estimator convention plus ≈0.06 dB compression at the operating drive; Appendix A). The same operator reproduces the solver's threshold to within 1.79×10^{-6} in M of its eight-seed, 240-round-trip median at N = 240 and cfl = 0.4 — the 0.11 % gap to the shipped single-seed, 90-round-trip value is the estimator's record length and seed, not nonlinearity — and to a largest sampled disagreement of 2.10×10^{-5} over fifteen (N, cfl) cells, a sampled maximum rather than a bound (demonstrated arithmetic; in-model as a statement about the nonlinear solver; notes/2026-08-07-mth-num-excess-decomposition.md); it also gives the effective round-trip amplitude exactly (l_a = 0.9110; the boundary loss is drift-induced — the passive cavity's boundaries are lossless in-model, so no scalar boundary factor exists). The exported +8 dB is therefore conservative for a design-length cell; the pre-registered expectations have now been run and resolved (notes/2026-07-12-predictions-resolved.md): the retuned solver run measures +9.22 dB (predicted +9.05 ± 0.30 — confirmed; the detuning artifact is empirically established, +1.43 dB recovered at identical bias, drive, and grid), M_th,num(N = 480) measures 0.158748 (predicted 0.1590 ± 0.0010 — confirmed, within 5×10^{-4} of the exact operator's linear value), and the half-bias check measures +5.13 dB (predicted +5.08 ± 0.30 — confirmed). Baselines were replicated bit-for-bit before any variant ran. Adoption of the retuned cavity length in the released chain is queued as WP1 work; until it lands, the shipped key retains the 7.80 dB configuration value. Near threshold the cell rings: the cavity ring-down reaches ~4–5 ps (comparable to the slot) and the pulse-to-pulse spread reaches 5 dB by 0.94·M_th. This is the regenerative amplifier's classic gain-versus-stability trade-off, and it is why the working point is held at 0.7·M_th — where the ring-down both lifts the per-pulse gain to its +8 dB streaming value and sets the inter-symbol budget the re-sync stations manage (Section 6.2). Every gain figure in this pass is a hydrodynamic-tier figure, and each carries a signed kinetic correction: −0.95 dB per cell at 353 K at the tracked operating point, band [−1.5, −0.3] dB, in-model within the closure of Section 7.5 (whether a 353 K tracked point exists at the kinetic tier is open there), plus a separate open ±1 dB boundary allowance (Section 7.5). None of the numbers above moves — the correction is stated on the ledger, not folded into it, and the tier that owns it has not run. It is worth recording why the correction is a decibel and not several: the tracked-ratio convention of Section 5 absorbs the kinetic shift of the boundary gain into bias current, and at a bias frozen at 0.7·M_th(hydro) the same physics would have cost −2.5 dB at the same temperature.

Because the operating drift sits far below v_sat(10^{12}) ≈ 1.4×10^{6} m/s, Joule heat falls ~25-fold against an at-v_sat drive at the same density, and ~150-fold against the Pass-1 corner. Of that ~150×, the dominant ~32× comes from the density itself (heat ∝ n·m*·v^2, with the effective mass m* = E_F/v_F^2 ∝ √n) and only ~5× from operating below the v_sat ≈ 5×10^{5} m/s of the high-density case. The exact periodic-slab thermal solution then gives the duty figures of Section 8.2. Heat stops selecting the density — disorder takes over that role: the chain's disorder Monte-Carlo (Figure 9D, reported with Pass 4) gave ~87–100 % cell yield at n = 10^{12} over a broad disorder sweep, degrading steadily toward 10^{11} as the logic swing

approaches the puddle scale; restricting to the literature disorder values (Section 7.4) tightens the n = 10^{12} yield to 98–100 %. Both are design-model yields, resting on the swing-to-puddle premise of Section 7.4.

Figure 8 — Pass 3: regenerative operating point (LF M_th,num ≈ 0.169; inviscid ≈0.149; kinetic 0.165)

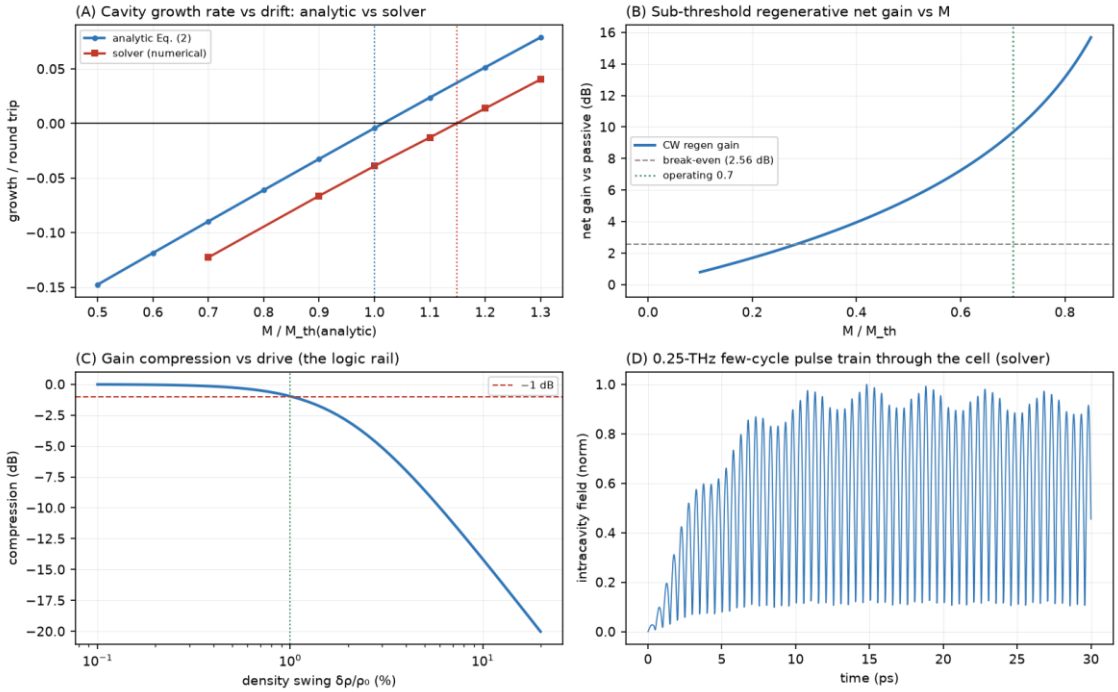


Figure 8. Pass 3 (coarse-grid Lax–Friedrichs M_th,num ≈ 0.169 — inviscid limit ≈0.149, kinetic threshold 0.165 within the closure of Section 7.5; operating ratio M/M_th,num = 0.7). *(A) Cavity growth rate vs drift, analytic Eq. (2) vs the nonlinear solver (computed at 70 round trips), both thresholds marked on the axis M/M_th(analytic): the analytic marker sits at 1.0 (Eq. (3)'s small-M 0.147) and the solver marker at the released 0.169 (≈1.15), ~15 % higher; the plotted Eq. (2) curve crosses zero at the exact inviscid 0.1493 (≈1.015), 1.5 % above the analytic marker, so about a tenth of the offset is the small-M expansion and the rest the scheme's numerics, chiefly bulk Lax–Friedrichs diffusion (Section 7.3). (B) Sub-threshold CW regenerative net gain vs M; break-even at the per-gate (λ_p/2) loss; +9.7 dB at the working ratio. (C) Gain compression vs drive amplitude — the compression knee (1 dB at ≈1 % swing; a compression point, not an output rail, Section 5). (D) The 0.25 THz few-cycle pulse train building up in the cell (solver, intracavity field).*

7.4 Pass 4 — cascade, noise, and measured disorder

The architecture's load-bearing claim — cascadability — was then budgeted at cell level: with cells chained through junction transmissions of 0/−1/−3/−6 dB, the analytic cascade budget — the continuous-wave cell transfer less its compression at the junction-attenuated operating swing — sustains +8.4 to +9.3 dB net per cell (two-cell total ~+17–19 dB over the passive chain, computed as twice the per-cell figure), with per-cell gain rising as junction loss moves the input down through and below the compression knee — the compression form's soft roll-off, not a rail: the budget is that form evaluated at the ledger swing A_op = 0.0116 (Appendix A), a constant that notes/2026-08-01-gated-requirement-round-trip.md recommends retiring as a ledger input, and the same form's own fan-out-2 fixed point is 0.0206 (0.0155–0.0237 over the A_SAT band), a knee-scale level (in-model; notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). The per-cell figure is the continuous-wave gain less ≈ 1.3 dB of in-model compression at the operating swing, edging

higher where junction loss moves the swing down toward and below the knee; its numerical proximity to the streaming figure of Section 7.3 (+8 dB) reflects those two conventions, not an independent cross-check. The direct two-cell time-domain test is open work, tracked in the project roadmap. These are hydrodynamic-tier figures: at 353 K the kinetic band of Section 7.5 reads the 0 dB-junction entry of this budget, 8.40 dB/cell, down to ≈ 7.45 dB/cell, band [6.9, 8.1], within its closure and before the open ±1 dB boundary allowance (whether a 353 K tracked point exists at the kinetic tier is open, Section 7.5). That entry is the only one the note maps, and the lossier junction classes are left uncorrected here rather than shifted with it: the per-cell figure is defined at the junction-attenuated swing, so its compression term differs from class to class, and a corrected per-class figure has to be recomputed against the cascade budget by the tier that owns the correction (Section 10). At the junction class the band does cover, the cascade still closes across the whole signed band; Section 6.3 records what the correction does to the fan-out budget, which is the margin that actually notices.

Noise. For a matched-temperature, phase-insensitive amplifier the noise-figure floor is F = 2 − 1/G (Eq. 7), i.e. 2.77 dB at the 353 K operating gain (+9.66 dB; the matched input floor cancels the thermal occupation, so temperature enters only through G(T); the Part-II decoder floor is the same closed form, with the promoted derivation note fixing a cross-manuscript check: at 353 K it must reproduce 2.768939660565078 dB to float precision, while below 150 K the two released lifetime models diverge by construction, 2.8012 vs 2.7925 dB at 77 K — a designed separation, not a discrepancy); total output noise then sits ≈ G·F ≈ +12 dB above the input thermal floor. This floor is what the re-sync spacing law of Section 6.2 consumes. Within Eq. (7) the bare cell's own noise figure stays below $10 \cdot \log_{10} 2$ = 3.01 dB at every gain G ≥ 1 (in-model), so the NF ≤ 4 dB requirement is met by construction, not by margin, and has content in two places only (notes/2026-08-05-idler-added-noise-and-the-nf4db-requirement.md). One is what precedes the gain: under the full-Friis upper bound, a blind flush of S ≳ 1.23 dB ahead of the gain takes the noise figure past 4 dB (in-model; notes/2026-07-22-flush-noise-figure-negative.md). The other is the amplifier's idler, whose bench form is one gain-free scalar: at the CW operating bias the cell's input-referred added noise must not exceed 11.14 quadrature units at 353 K (9.47 at 300 K) on this section's matched floor $\bar{n}$ + ½, or 9.31 (7.80) referred to the variance Part II's comparator receives — a figure that holds at that chain's 1-dB junction loss without a preamplifier, whereas the matched-floor row is free of such parameters and 19.6–21.4 % more permissive — so the figure must be quoted with its floor (demonstrated). All of this is conditional on booking the regenerative buildup as the amplifier gain, which the released chain does not settle (open). An absolute added-noise calibration against this floor — the one quantity the reduced-order chain bounds but does not pin — is assigned to the Boltzmann–Maxwell tier (Section 10). The kinetic band of Section 7.5 barely touches the floor itself: at the corrected 353 K gain F reads 2.71 dB rather than 2.77 dB, since near +9.7 dB the floor moves only ≈ 0.06 dB per dB of gain. It does, however, add a dissipative channel, and whether that channel adds noise at the fluctuation-dissipation rate is not something the closed form answers — it belongs to the same absolute calibration, and is open with it.

$$F = 2 - 1/G \qquad (7)$$

Finally, literature disorder values ($\sigma \approx 2.5\times10^{9}$ cm^{-2} for graphene on ultra-flat hBN [12]; $\sim4\times10^{10}$ cm^{-2} for typical encapsulated stacks [14], with SiO_2-substrate values an order higher as the upper bound [13]) place cell yield at $n = 10^{12}$ at 98–100 % across the realistic range; 3×10^{11} is a premium-material option (graphite-gate class). The yield criterion is explicit: a charge puddle of rms density σ broadens the cell's threshold to $\sqrt{((n/k)^2 + \sigma^2)}$, so the puddle-broadened sharpness clears the regeneration floor only while σ stays below ≈ 0.16·n — the swing-to-puddle rule, tied directly to the k ≥ 8 design sharpness (Appendix A). The rule inherits disorder.py's premise β·A_swing = n, that is, k = 8 at zero puddle. Executed on the retuned cavity at the absolute bias, a delivered '1' shifts the released cell's drain-plane density by only 0.017–0.021·n, so the premise is ≈ 50× off and the released cell's zero-puddle sharpness (0.18–0.26 on the tanh scale) is already below the regeneration floor; the derivation applies to no released cell, and the 98–100 % yield above, with the density selection argued from it, is a design-model result resting on that premise (in-model; notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md).

Figure 9 — Pass 4: cascade, noise, and measured disorder

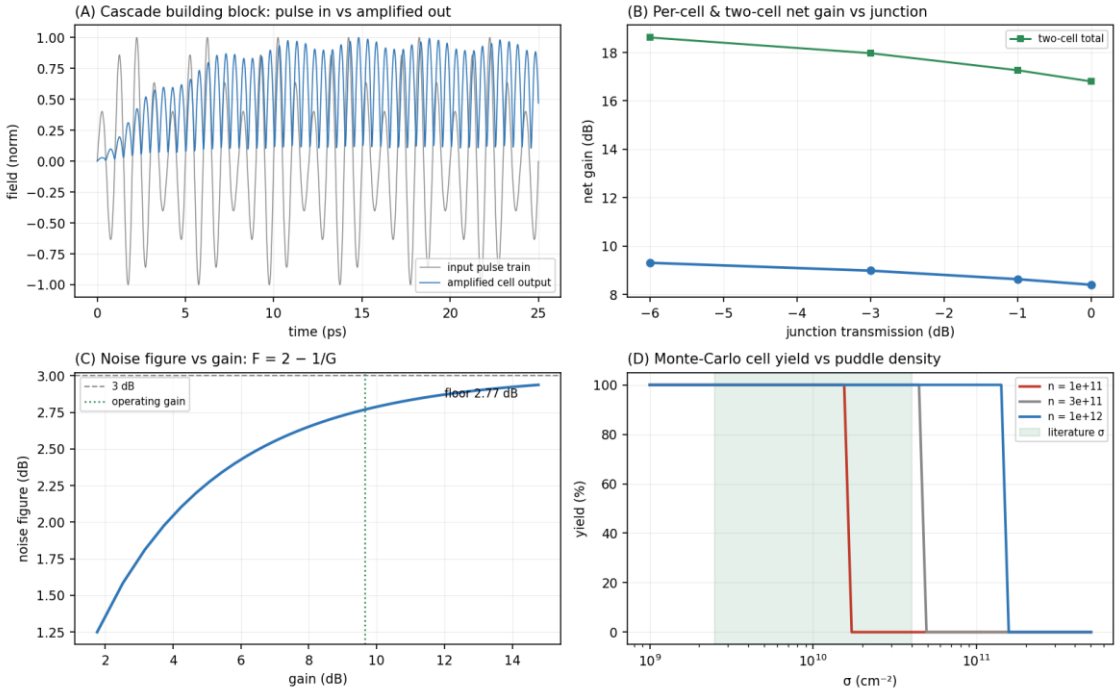


Figure 9. *Pass 4. (A) Cascade building block: the input pulse train and the amplified cell output (solver). (B) Per-cell and two-cell net gain vs junction transmission — per-cell gain rises as junction loss moves the input down through and below the compression knee. (C) Noise figure F = 2 − 1/G vs gain; floor 2.77 dB (353 K) at the operating gain. (D) Monte-Carlo cell yield vs puddle density σ, with the literature range shaded; n = 10^{12} clears it with margin on the design premise k = 8 (Section 7.4).*

7.5 Pass 5 — validity bounds and the last parasitic channel

Kinetic validity. The hydrodynamic equations behind Eq. (2) and the Pass-3 solver assume electron-electron collisions fast enough to keep the electron system a local fluid — quantified by $\omega\tau_{ee} \ll 1$. The standard estimate for graphene's electron-electron rate, $1/\tau_{ee} \approx (k_BT)^2/(\hbar E_F)$ [15], gives $\tau_{ee} \approx 115$ fs at 300 K and $E_F = 117$ meV — i.e. $\omega\tau_{ee} \approx 0.7$ at 1 THz. Two qualifications belong with that number. Its rate prefactor is uncertain

by a factor of a few — the Fermi-liquid log, screening and coupling-constant spreads, and the fact that the stress-relaxation time the wave actually samples is not the quasiparticle lifetime (measured graphene viscosities sit inside that band; A below denotes the multiplier on this prefactor, admissible band 0.5–3, and the relaxation rate of the current not locked to momentum is a separate, open quantity, Section 10). And the group is temperature-dependent in the design's favour: with $\tau_{ee} \propto T^{-2}$, at the 353 K cap $\tau_{ee} \approx 83$ fs and $\omega\tau_{ee} \approx 0.52$, so hydrodynamics is better at the hot end, which is where the fabric runs. The honest statement is that the hydrodynamic description at 1 THz and 300 K is marginal, not comfortable: $\omega\tau_{ee} \approx 0.7$ sits at the collisional-to-collisionless crossover — the same crossover observed experimentally as the transition from acoustic plasmon to electronic sound [10] — with $\ell_{ee}/L \approx 0.2$ (0.14 at the cap, where the sharper wave-kinetic Knudsen numbers are $q_-\cdot\ell_{ee} = 0.25$ and $q_+\cdot\ell_{ee} = 0.20$), and the viscous correction νq^2 ($\nu = v_F^2\tau_{ee}/4$) is ≈ 20 % of $1/\tau$ — a 300 K figure that reads 0.13 at the cap, and that overstates the actual dissipative supplement in the conservative direction, since the collisionless crossover reduces it by $1/(1 + (\omega\tau_{ee})^2)$ to ≈ 0.10 of $1/\tau$ there. The cell is marginal in space as well: $L/\ell_{mfp} \approx 0.6$–0.7 with $\ell_{mfp} = v_F\cdot\tau$ the momentum-relaxation mean free path, so transport in the cell is neither cleanly hydrodynamic nor cleanly ballistic. The gain numbers of Passes 3–4 therefore carry a kinetic correction, and its certification passes to the Boltzmann–Maxwell tier as that tier's highest-priority question (Section 10) — but the correction is signed. A community note promoted by a 3-of-3 agent vote (notes/2026-07-13-kinetic-correction-signed-band.md, July 2026) closes the system one tier up in the Maxwell/Mermin viscoelastic generalization of the shallow-water equations, $\nu(\omega) = \nu_0/(1 - i\omega\tau_{ee})$ with $\nu_0 = v_F^2\tau_{ee}/4$ — a closure exact at the collisionless endpoint and, at the hydrodynamic one, within a 0.12 % speed offset of the exact $T \to 0$ Dirac-fluid limit (its zero-to-first-sound shift is ¼ v_F^2 against the exact 0.2627 v_F^2; in-model, notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md), reducing to Navier–Stokes with the chain's own ν_0 as $\omega\tau_{ee} \to 0$ and the collisionless mode as $\omega\tau_{ee} \to \infty$ — and returns four results at the operating point. First, single-particle Landau damping at the operating q is not merely avoided but kinematically excluded, with and without drift: intraband particle-hole pairs on a Dirac cone obey $|\omega| \le v_F q$ identically, a property of the spectrum and not of the occupation, so $\mathrm{Im}\,\Pi_{intra}$ vanishes above v_F for any electron distribution — drifted, rigidly shifted, boosted, or thermally broadened to any temperature. The factor ≈ 2.3 of Section 5 is a margin on an identity, not on an estimate, and the counter-propagating leg — the one carrying the larger wavenumber — still retains a factor ≈ 2.1. This is the chain's one demonstrated kinetic result; thermally unblocked interband absorption at $\hbar\omega \ll 2E_F$ adds < 0.001 dB beside it. Second, the residual kinetic damping is signed and negative: at 353 K and the tracked $M/M_{th} = 0.7$ the per-cell gain correction is −0.95 dB, band [−1.5, −0.3] dB, with the threshold, within the closure, shifting to $M_{th} \approx 0.165$ — ×1.12 on the analytic 0.147083, band ×[1.05, 1.15]. At 300 K the same closure gives −1.27 dB, band [−1.6, −0.5] dB, with the threshold ×1.16. Both figures are statements within the closure. Run with a single relaxation rate in the finite-temperature bulk, with the DS clamps on the plasmon pair and the base at the lattice temperature, the kinetic operator Section 10 specifies (number-, momentum- and energy-conserving relaxation about the drifted equilibrium) carries ×3.8 the closure's electron-electron damping at 353 K (0.387 against 0.101 of $1/(2\tau)$ at A = 1), about two-thirds of it in the relaxation of the current not locked to momentum, whose rate the prefactor band does not

bound. When the electrons shed their drift heat (energy relaxation ≤ 0.3 ps) it finds no DS threshold at 353 K for A ≤ 1 below the drift at which the backward leg's phase speed falls to v_F, and at 300 K one only at A = 3; where it finds none the tracked correction is undefined rather than shifted. The smallest lifetime factor that restores a 353 K threshold there is ×2.56–2.60, to 2.18–2.21 ps against 0.85 ps (a bisection bracket whose lower end rests on the DC-drive closure); neither a ×6 higher f_0 nor the d = 5 nm die restores one. That run is not the Section 10 solve, so the band is not falsified, and whether graphene's own electron fluid has a 353 K threshold is open: three admissible two-rate continuations, identical at zero drift, give three answers, two of them restoring a threshold with no change of τ (in-model; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md). The sign survives the prefactor uncertainty on $1/\tau_{ee}$, because the damping supplement is positive-definite for every τ_{ee} and has a bounded interior maximum at $\omega\tau_{ee} = 1$: a factor-of-a-few error bounds the correction's magnitude rather than flipping its sign. The band is in-model at the closure's own level — its ±25 % crossover allowance is judgment, not theorem, and the map from the cavity pole shift to a per-cell gain carries a ±0.2 dB translation error, inside which a 0.5-ps transit-window convention is disclosed but ad hoc and awaits pinning to the chain's own gain definition. The correction is a decibel rather than several only because the operating point is a tracked ratio (Section 5): with M_{th} calibrated in situ, any multiplicative renormalization of the DS boundary gain is absorbed into bias current and leaves the tracked loop gain alone to better than 0.04 dB per ±20 %, so only additive damping and detuning move the gain. At a bias frozen at $0.7 \cdot M_{th}(\text{hydro})$ the same 353 K correction would read −2.5 dB. Third, the mode runs slower than the design value: Eq. (1) is the collisionless endpoint, and at the operating $\omega\tau_{ee}$ the in-closure speed deficit is −1.8 % at 353 K, band [−2.3, −1.2] %, detuning the fixed-length cavity against a model-derived 2.8 % half-linewidth at a further cost of ≈ −1.5 dB at the central deficit (−0.8 to −2.3 dB across the band) if left uncorrected. A density trim of $\Delta n/n \approx +7$ % nulls it; the gain band above assumes that trim, which is therefore an explicit design requirement rather than an implication of the calibration philosophy, and is measurable at bench gate G1 (Section 10). Finite-temperature compressibility corrections to s at $T/T_F = 0.26$ lie outside the closure, whose own band on the total, [−4, +1] %, contains zero; only the in-closure part is signed there. Computed in-model in the single-rate kinetic operator that Section 10 specifies, run in the bulk, the bulk mode at 353 K sits 11.1 % below Eq. (1) at A = 1 (8.8–12.6 % over the A band; 6.0–12.7 % across two-rate variants), of which about 9.4 % is the finite-temperature part once the in-closure 1.8 % is removed, and the zero-drift cavity line sits 11.7–12.0 % below the released model's continuum line (band 9.6–13.4 %), outside the closure's band. The density trim that restores Eq. (1)'s speed in that kinetic bulk is +47.6 % (band +36.5 to +56.5 %), not +7 %. The bench value stays open, and the pre-registered in-closure target of Section 10 stands (notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md). Fourth, and this is what the next tier — or one cheap bench measurement — is for: an additional ±1 dB allowance rides on top of the band, owned by kinetic boundary layers at the clamped contacts (ℓ_{ee} ≈ 83 nm at 353 K, ≈ 14 % of the cell, at each end). It does not close on paper: the boundary loss δ_b is a sign-definite loss at the density-clamped source, sign-open at the DS-active drain, and set entirely by the contact accommodation — with even its Knudsen-layer scaling exponent open — so the honest centre is a net loss and the allowance could be larger than ±1 dB, not smaller — a diffuse edge costs ≈−3 dB under a Kn^2 law and ≲−10 dB under a Kn law. Its

sensitivity is −0.28 dB per 1 % of additive round-trip loss. What analysis can do is make it falsifiable at the bench: because δ_b is a per-reflection loss while the bulk viscous channel is distributed, three sub-threshold cavity-linewidth signatures separate them — a contact-preparation contrast isolates δ_b (≈3.2 GHz of excess half-linewidth per 1 %), a temperature sweep reads its Knudsen exponent, and a drift sweep reads the drain sign — so a near-model linewidth on a specular-contact cell gates an architecture-led Part III without the full solve, and a broad line measures the three inputs the solve otherwise owes (Section 10). Mapped onto the corrected ledger at 353 K the band reads: continuous-wave 9.661 dB → ≈ 8.7 [8.2–9.4] dB; retuned streaming 9.2246 dB → ≈ 8.3 [7.7–8.9] dB; cascade 8.40 dB/cell at the 0 dB junction → ≈ 7.45 [6.9–8.1] dB/cell. The ledger values themselves do not move — they are what the released chain computes, and this is a stated correction on them, not a rewrite of them — but two margins do, and are recorded where they arise: the thermal envelope of Section 8.2, because the threshold shift is real bias current, and the fan-out budget of Section 6.3. The note also audits Pass 5's own validity keys: the shipped ωτ_ee = 0.722 is a 300 K number and 0.52 belongs beside it at the cap; viscous_fraction = 0.209 is likewise 300 K, omits the crossover reduction, and — the substantive finding — was never propagated into gain, the propagation being what this paragraph now reports, and it, not the raw fraction, is the design-relevant number; and hydro_expiry_K = 254.9 is retired as a concept in Section 8.3.

Coupled-cell stability. Two DS-active segments on one continuous channel, with a midpoint density clamp swept from hard contact to fully free, are stable at every coupling strength, even at 0.9·M_th — the DS increment's 1/L scaling protects long parasitic cavities (Figure 10). The check is a closed-form ansatz, not an eigenvalue solve: the released Pass 5 assumes an effective threshold M_th·(2 − c) across the clamp strength c, whose growth rate is largest at the hard-clamped, independent-cell end, so stability at every clamp strength follows from the sub-threshold bias alone, while the free end (one 2L cavity) is Eq. (2)'s 1/L scaling (in-model; source read of coupled.py). Whether a coupled two-segment operator has a supermode below the single-cell threshold is open, so charge-channel parasitics are excluded only under that ansatz; beyond it, the remaining coupling channel is radiative/electromagnetic, which is precisely scoped to the Boltzmann–Maxwell tier.

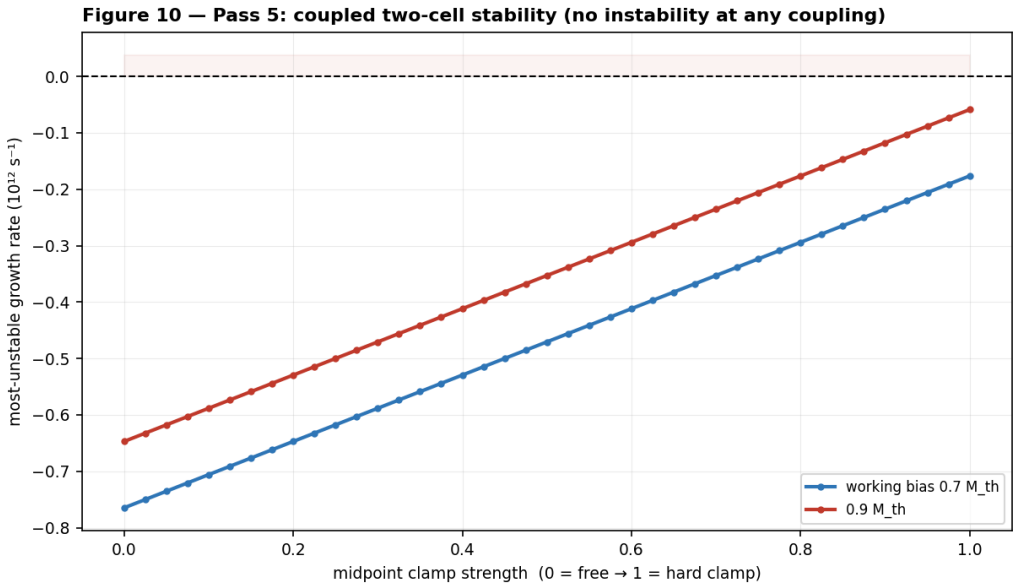


Figure 10. *Pass 5: most-unstable growth rate of two coupled DS cells vs midpoint clamp strength under the closed-form coupling ansatz of Section 7.5, at the working bias and at 0.9·M_th — no instability region is entered at any coupling.*

7.6 Scoreboard

Table 6 collects where the five passes leave each of the chain's questions.

Question	Status after five passes
Net gain ≥ loss at 1 THz, ≤ 80 °C	in-model yes: regenerative cell, ~+8 dB/pulse streaming, per-slot-peak metric (CW +9.7 dB) at M/M_th = 0.7; calibration budget in Section 7.3
Cascadability	in-model yes: 2 cells, +8.4–9.3 dB/cell through −6 dB junctions (compression-form budget at A_op; levels re-clamp near a 2 % drive, Section 5)
Noise	NF floor 2.77 dB at 353 K (F = 2 − 1/G); ≤ 4 dB required — met by the bare cell at any gain, so it binds on what precedes the gain and on the idler (bench scalar, Section 7.4); absolute calibration to the Maxwell tier
Disorder	98–100 % yield at n = 10^{12} for literature σ; swing-to-puddle criterion (σ ≲ 0.16 n); measured on-die in the protocol; a design-model result on the premise k = 8, which the released cell does not supply (Section 7.4)
Thermal	~23 % margin at worst-case fill ⅓ (hydrodynamic tier; ≈ 4 % within the closure of Section 7.5; in graphene's own fluid, every computed row with a threshold exceeds the budget under that fluid's own Joule form J·E_0, in-model); thin (Section 8.2)
Kinetic validity	marginal (ωτ_ee ≈ 0.7): signed kinetic band −0.95 dB/cell at 353 K [−1.5, −0.3] dB, plus an open ±1 dB contact-boundary allowance — sign-open at the drain, loss-definite at the source, potentially larger for a diffuse contact; the band is the closure's, and whether a 353 K tracked point exists at the kinetic tier is open (the single-rate kinetic operator finds no

	353 K threshold for A ≤ 1 when the electrons shed their drift heat); passes to the Boltzmann–Maxwell tier (Section 7.5)
Coupled-cell parasitics	charge channel stable at all couplings under a closed-form coupling ansatz (a coupled-operator supermode is open); radiative channel open
Clock synthesis	CW chain demonstrated; pulsed multi-line extension proposed, undemonstrated (Section 3.2); source class and distribution budget open (issue #139)
Launch/read-out coupling	efficiency into the acoustic mode and the launcher's amplitude jitter at the launch plane unquantified; open (Sections 3.3, 9)
Remaining	Boltzmann–Maxwell solve (first, whether a 353 K threshold exists; then the kinetic gain, with EM coupling and absolute calibration in the electromagnetic tier) and the bench (Section 10)

Table 6. *The feasibility scoreboard: the ten questions the model chain was built to answer, and where each stands.*

8. Energy accounting and thermal envelope

8.1 Energy

The intrinsic field energy is ~10 aJ per bit. The drive energy is set by the bias: at the operating point the per-cell dissipation is ~1.2 kW/cm^2 ($P = n \cdot m^* \cdot v_0^2/\tau$ with $m^* = E_F/v_F^2$, $v_0 \approx 2.4\times10^5$ m/s, $\tau = 0.85$ ps), rising to ~1.7 kW/cm^2 if the in-situ threshold calibration lands at the solver-class bias (Section 7.3). Both are hydrodynamic-tier figures at the 80 °C cap, computed at the hydrodynamic threshold; Section 8.2 records what the kinetic threshold shift of Section 7.5 does to them, and why it does not move either number here. For the half-adder block — seven biased cells (the five logic cells plus the two carry-alignment buffers, counted conservatively) of ~10^{-8} cm^2 each — that is 0.09–0.12 mW, i.e. ≈ 0.35–0.5 fJ per addition at one addition per 4 ps, the design-baseline slot — CMOS-class, not below it. The bias is DC and the released chain books energy per addition as block power × slot time, so the figure grows in proportion to the slot at the slower rate where fan-out-2 logic closes in-model (Section 6.2); Part III books its lane at the 69-GHz gated slot [45]. Neither dissipation figure books contact Joule heat: if each cell's ~100 Ω·µm-class source contact (Section 9.1) carries the row's series bias current (Section 9.2), it adds J^2R_c ≈ 14.8 µW per µm of width at the hydrodynamic-tier bias at the cap, about twice the channel's own 7.3 µW/µm (in-model arithmetic on the released bias current and dissipation); sunk through the same thermal path, it would exceed the whole margin of Section 8.2. Whether the contact carries the bias current, and how its heat is sunk, is open.

The system energy — microcomb wall-plug, DFB lasers, photomixers, detectors — is far larger. A 1 W-class drive over 10^4–10^5 active cells at the 0.25 THz baseline symbol rate corresponds to ~40–400 aJ per operation (20–200 aJ at 0.5 THz). Any "orders below CMOS" claim survives only for the intrinsic field energy, never for the system [16, 17].

8.2 Thermal envelope

The ≤ 80 °C cap is a 55 K rise budget over 25 °C ambient, against an effective stack resistance $R_{th} \approx 0.10$ K·cm^2/W (consistent with ~500 µm of sapphire plus interface terms; the same R_{th} reproduces every duty figure in this manuscript). At the worst-case area fill of one-third, continuous drive at the operating bias gives $\Delta T \approx 42$ K — the budget closes with ~23 % margin. The margin is real but thin: at the upper, solver-class calibration bound of Section 8.1 the same calculation gives $\Delta T \approx 55$ K — the budget exactly saturated, zero margin, before any duty throttling. The budget therefore closes at the operating bias and sits right at the edge at the upper calibration bound; the thermal row of the scoreboard (Table 6) says both. That thin margin thins further under the kinetic correction of Section 7.5, and this is the correction's sharpest architectural consequence. The tracked bias hides the kinetic threshold shift from the gain, but not from the bias supply: at the 80 °C cap, tracking to a threshold ×1.12 higher costs $(M_{th,kin}/M_{th})^2$ in dissipation — ×1.25 centrally, ×1.32 at the $\omega\tau_{ee} = 1$ bound — so the operating-bias figure of Section 8.1 becomes ≈ 1.59 kW/cm^2 (1.67 at the bound), the drift current density of Section 9.2 becomes ≈ 0.43 mA/µm, and, at the worst-case fill of one-third, $\Delta T \approx 52.9$ K centrally and 55.5 K at the kinetic worst case, against the same 55 K budget — all in-model. The ~23 % margin is a hydrodynamic-tier figure and reads ≈ 4 % centrally and zero at the kinetic worst case within the closure of Section 7.5, in-model, before any duty throttling. No number the released chain produces changes: the kinetic worst case coincides numerically with the upper, solver-class calibration bound of Section 8.1 (~1.7 kW/cm^2), whose $\Delta T \approx$ 55 K this section already quotes, so the envelope above already contains it. What changes is the reading — and it makes the Boltzmann–Maxwell solve of Section 10 the arbiter of the thermal budget as much as of the gain. The ≈ 4 % margin is the closure's. In every computed row of graphene's own electron fluid that has a threshold, tracking 0.7 of it costs ×1.2–2.1 the Section 8.1 dissipation in the released Joule form for the hydrodynamic closures and ×3.2 for the one kinetic row (300 K, A = 3), giving ΔT at one-third fill of 49–89 K and 134 K — inside the budget only for one 300 K hydrodynamic closure (49 K); graphene's own $J \cdot E_0$ adds a further ×1.25–1.35 (61–120 K and 176 K), and every such row then exceeds the 55 K budget. At 353 K the single-rate kinetic operator gives no tracked point to price (in-model; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md).

The half-adder block itself, at ~4 % fill, sits at $\Delta T \approx 5$ K and is thermally unconstrained. The positive-feedback loop ($T \to \tau \to Q \to$ loss → more drive) is defanged by the ~150× headroom between the operating dissipation and the at-v_{sat}, high-density corner of Pass 1 (whose own duty ceiling, ~0.3 % at full fill or ~0.9 % at fill ⅓, is fatal either way). Local heat flux at the cell (12–17 MW/m^2) is benign against demonstrated hBN/substrate interface conductances; the budget is set by the substrate spreading term, not the interface (Section 9.7).

8.3 Behaviour under cooling — and the validity window

The cap also has a cold side. Re-running the Pass-3 cell with the $\tau(T) = \tau(300\text{ K})\cdot(300/T)$ loss model shows per-gate loss falling from 2.17 dB at 300 K to 1.45 dB at 200 K (per $\lambda_p/2$; Appendix A), the thermal noise floor dropping with T, and the servo-biased cell remaining regeneratively stable at every temperature swept. That stability is conditional on bias

tracking: M_th scales with T (Eq. 3), so a bias fixed at 0.7·M_th(353 K) crosses the ∼0.8·M_th ringing onset after only ∼44 K of cooling — for an untracked bias, cooling, not heating, is the unstable direction.

The certifiable window, however, is set by model validity, and it is not a cliff. The chain ships a key hydro_expiry_K = 254.9 — the temperature at which ωτ_ee reaches 1 under the unit rate prefactor — but neither the number nor the concept behind it certifies anything. The number inherits the rate prefactor as $255/\sqrt{A}$: across the admissible band it spans 147–360 K, a range that contains the design temperature, so as a safety statement it is empty in both directions. And ωτ_ee = 1 is not an expiry but the point where the closure's damping supplement is largest at a given temperature and bounded there: at each temperature the tracked penalty has a floor over the prefactor band, ≥ −1.2 dB at 353 K and ≥ −1.6 dB at 255 K, before the closure's own ±25 % allowance and before the open ±1 dB boundary allowance — while the mode itself demonstrably survives the crossover, which is exactly what the acoustic-plasmon-to-electronic-sound experiment [10] shows. What replaces the expiry picture is a smooth, signed penalty that deepens as the fabric cools: in-model and tracked, −0.95 dB per cell at 353 K, −1.27 dB at 300 K, −1.55 dB at 255 K. That ladder is the closure's; in the single-rate kinetic operator, with the electrons shedding their drift heat, there is no 353 K threshold to track for A ≤ 1, and no driven transfer was computed (in-model; Section 7.5). Below 255 K the closure has not been evaluated, so the 200 K end of the sweep above carries no kinetic figure at all — that end is open, and the 255 K number bounds nothing below itself. What the hydrodynamic chain certifies is the loss trend, the tracked-bias requirement, and the absence of any predicted instability under cooling with a tracked bias; what it does not certify — the absolute gain, at any temperature — passes to the Boltzmann–Maxwell tier with the band of Section 7.5 attached, deeper at the cold end than at the cap. (Below ∼150 K the 1/T phonon scaling is itself optimistic — impurity scattering saturates τ.)

9. Device implementation considerations

This section collects what a fabrication-and-measurement team would need that the architecture sections do not state. Each item is either specified, bounded, or explicitly handed to the bench protocol or the Boltzmann–Maxwell solve.

9.1 Boundary conditions and contacts. The DS asymmetry is electrical, not geometric: density-clamped means AC-shorted gate-to-channel voltage at the source end; current-clamped means AC-open at the drain end [11, 19]. The practical realization is a low-impedance ohmic source contact (one-dimensional edge contacts at ∼100 Ω·μm-class resistance [33]) under the gated region's edge, and a drain side terminated by the gate-segment boundary into a depleted (CNP) region. No experiment has realized verified DS boundary conditions at the 576-nm scale; the design's protection is that M_th is calibrated in situ by approach to self-oscillation, which absorbs boundary-condition imperfection into the measured threshold to first order — provided the cell has a threshold to approach, which at 353 K is open at the kinetic tier (Section 10). Residual non-ideality appears as a modified increment and is exactly what bench gate G1 measures. The kinetic note of Section 7.5 sharpens what that protection covers and what it does not. It covers the multiplicative part, and covers it well: any renormalization of the boundary-gain slope

moves the threshold — bias current, dissipation — but leaves the tracked loop gain alone to better than 0.04 dB per ±20 %, which is why a defensible band exists at this order at all. It does not cover the additive part. Electron-electron collisions are slow enough at this operating point that a Knudsen layer of ℓ_ee ≈ 83 nm at the 80 °C cap — about 14 % of the cell — sits at each clamp, and whatever dissipation those layers add is charged straight to the gain at −0.28 dB per 1 % of additive round-trip loss. It does not close on paper: it is a sign-definite loss at the density-clamped source and sign-open only at the DS-active drain (whose reflection could be kinetically enhanced as well as degraded), so the honest centre is a net loss. That ±1 dB allowance is the largest open kinetic exposure of the gain band — whether a 353 K threshold exists at all comes before it and is owned by collision-operator fidelity (Section 10) — and it is why contact fidelity leads the Boltzmann–Maxwell tier's ingredient ranking for that band (Section 10), and it makes the specular-versus-diffuse character of the clamps a device-physics question the bench should not be the first to answer.

9.2 Bias network and current density. The drift current density at the operating bias is $J = n \cdot e \cdot v_0 \approx 0.38$ mA/μm at the 80 °C cap — roughly an order of magnitude below the current-carrying capacity demonstrated for encapsulated graphene, so electromigration and channel burnout are not at risk at the operating point. That is the hydrodynamic-tier value, computed at the hydrodynamic threshold; tracking the kinetically corrected threshold of Section 7.5's closure raises it to ≈ 0.43 mA/μm in-model (Section 8.2), which changes the conclusion in neither direction — the margin to the current-carrying capacity remains an order of magnitude. In the computed rows of graphene's own electron fluid that have a threshold, the tracked current density is higher still — 0.50–0.56 mA/μm in the isothermal hydrodynamic closures at 353 K with the drift heat shed (0.45–0.50 mA/μm at 300 K), and 0.74 mA/μm for the one kinetic row (300 K, A = 3, projected DC drive) — while at 353 K the single-rate kinetic operator gives no tracked point (in-model; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md); the order-of-magnitude margin is stated for 0.38–0.43 mA/μm and is not re-evaluated at these values. Rows of cells in series share one bias current; parallel rows need per-row ballast so that a low-resistance row does not starve its neighbours. The bias envelope (kHz–MHz) is far inside the bias-plane RC corner.

9.3 Gate-lattice programming. The program is a static voltage map on ~576-nm-pitch top-gate segments, written by an electrode-mapping controller at kHz–MHz rates — orders of magnitude below the gate-line RC limit for millimetre-scale lines, so routing in two perimeter metal layers suffices. Fringing-field crosstalk between adjacent segments at d = 10 nm is at the percent level of the programmed density step and below the disorder scale the yield analysis already absorbs.

9.4 Inter-cell junctions. A junction is a gate-defined density step between adjacent segments. For the ideal step the velocity mismatch is mild ($s \propto n^{1/4}$-class under Eq. (1)), and the reflective loss of even a 2× density step is < 0.1 dB — the −1 to −3 dB budgeted per junction is therefore dominated by non-ideal effects: scattering at the gate gap, mode mismatch through the ungated sliver, and radiation. Those belong to the Boltzmann–Maxwell tier; the bench die carries junction variants to measure them directly. The

budget's robustness to the answer is already established in-model on the compression-form cascade budget (it closes through −6 dB junctions, Section 7.4; no multi-cell cascade has been run).

9.5 Launch and read-out transducer. The unsolved link of the chain (Sections 3.2–3.3): the photomixer and stripline modes are micron-scale while the acoustic plasmon is confined ∼100× tighter, and no cited work quantifies the coupling efficiency. With μW-class launch power at 1 THz (∼10 aJ per slot) the budget tolerates little coupling loss. Mitigations in order of maturity: the photoconductive launch demonstrated on this stack at 4 K [4]; tapered-gate impedance transformers; and a resonant antenna-coupled gate segment. On the read-out side, photoconductive sampling is field-resolved and coherent, and the in-situ reference arm [7] removes absolute-calibration error; an InP HEMT THz low-noise amplifier (LNA; 9 dB at 1 THz [28]) inserted before detection is an optional hybrid that relaxes the detector requirement by an order of magnitude. That relief is against the detector's own noise only. What the read-out can resolve is set by the swing the port receives as well as by the detector, and only the input-referred swing counts: an amplifier ahead of detection cannot raise the input-referred signal-to-noise ratio (the released amplifier multiplies it by $1/(2 - 1/G) < 1$) and does not lower the decision error's quantum floor. In-model, a homodyne-limited decision (field-resolved, no amplifier noise, midpoint threshold) at Part III's digital pass line of 6.3×10^{-9} per decision [45] needs ≥ 406 quanta at 300 K (477 at 353 K) — above the fan-out-2 cascade's fixed point (99–277 quanta) but below the fan-out-1 level (628–955, a level outside the released model's velocity domain) and the launch budget (576–737 quanta, a budget rather than a delivery); which swing the read-out port receives, the loaded cascade's or the launcher's, is open (notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md; issue #141). Coupling efficiency joins the calibration sequence as a measured quantity (Section 10), and so does the launcher's pulse-to-pulse amplitude jitter σ_{rel} ($\sigma_A/\bar{A}$ of the launched '1') at the in-fabric, post-coupling launch plane. Which launcher physics and reference plane the chain provides — additive at full swing, proportional, or rail-referenced — decides whether the 20 dB of Section 6.2 is a thermal link budget or a launcher self-noise specification, and it spans ×239 in Part II's decode error and ×12 in station spacing; a pre-coupling bench measurement does not transfer until the coupling side is fixed (open; notes/2026-08-25-what-the-20db-launch-budget-denominates.md).

9.6 Re-sync station hardware. A re-sync station is a DS cell used as a clocked AND: its bias gate is driven by the optical clock tap through a photoconductive element beside the top-gate segment, so the cell has gain only during clock slots. The tap inherits the comb's phase (Section 3.2 budget), needs only enough optical energy to swing the local n_{bias} across n_{th} (fJ-class at 10-fF-class segment capacitance), and doubles as the ASE flush of Section 6.2. The tap must swing the gate within each slot — for 10 dB of extinction about $-0.20\cdot n$ on the drain density (Section 5) — and the gate's RC at that speed is not modeled by the released chain; whether the tap can double as the electrical stage of a gate-segment inverter within the slot is open (notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md), and issue #139 asks whether the clock's burst modulator rests on the same element. Whether gating the bias also re-references the carrier phase of the pulse the station passes, outside the clocked regime of Eq. (5), is open too: on it depends whether stations bound the run of

cells over which a per-cell phase error accumulates — if they do, the station cuts of Part III's lane (every 10 or 13 cells [45]) cap that run at 9 or 12 cells, ±0.11 % or ±0.08 % of f_0 per cell at the longest (in-model; notes/2026-10-13-phase-is-not-the-easier-half-for-the-cell.md).

9.7 Tolerances and thermal interfaces. Cell-length error of ±5 % (±29 nm — loose for deep-UV overlay) detunes f_0 by ∓ 5 %, which lies outside the biased cell's line rather than inside it: in-model the active half-width is 2.76 % of f_0 (27.6 GHz, Section 7.5's 2.8 %) against the passive line's 9.4 % half-width, and the error moves the cell's transfer phase by +85° (5 % shorter) / −34° (5 % longer) in the shipped configuration, 2–6× the ±15° budget of Section 3.2, while a +5 % drive detuning costs 6.5–7.0 dB of delivered amplitude (notes/2026-10-13-phase-is-not-the-easier-half-for-the-cell.md). hBN thickness is layer-quantized, and a ±1-layer error on d = 10 nm shifts s by under 2 %. Both are absorbed by per-cell top-gate trim against the in-situ reference, which makes trim resolution a requirement: ±15° at one cell is −5.0 nm (shorter) / +7.3 nm (longer) of length (2.7–10 nm across the configurations), finer in a cascade, and the untrimmed −1.8 % speed deficit of Section 7.5 alone moves the phase by about −20° (in-model). The local heat flux (12–17 MW/m^2) drops ~K-class temperature across demonstrated hBN/substrate interfaces; the thermal budget is set by substrate spreading (Section 8.2), so die layout should interleave active stripes with dark guard rows, which the cell map already does.

9.8 Hybrid integration: the device combination. The minimal bench die combines three device families: the fiber-bonded photonic chain (microcomb, DFB pair, UTC-PD [5]) at the edge; the photoconductive launch/read-out circuit on the substrate [7]; and the graphene fabric itself. The full machine adds two more: a conventional-CMOS perimeter layer for charge memory, refresh, trim DACs and the electrode-mapping controller (Section 6.4); and, optionally, room-temperature THz electronics as accelerators — an InP HEMT LNA before read-out [28], and RTD latches as fast electronic registers or an injection-locked local clock auxiliary [29]. If an inversion uses the gate-segment route of Section 5 — a detector segment's rectified output amplified onto the inverting cell's gate within ≈ 7 ps and recovering within the 2.9-ps gap between bursts, ×9–12 behind a biased detector for a −0.20·n swing (×8–17 across the −0.17 … −0.28·n band where 10 dB is reached; in-model, the gate's RC open; notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md) — the machine needs a further member that no Part specifies: an electrical gain stage at each such inverter. Its device class and placement, and the class of the clock source, are open design questions, tracked as issues #141 and #139. Every member of this combination is room-temperature hardware; the integration, not any component, is the proposal.

10. Falsifiability: the experiment and the solve

The bench protocol is a lab-ready measurement plan. It fixes in advance the device, the drive and read-out chain, the calibration sequence, and the pass/fail gates. Each threshold is parameterized by the model predictions, so a large deviation is itself a result that recalibrates the model chain rather than a tuning failure.

The device under test: an hBN-encapsulated dual-gated DS cell on the Section 3.1 stack, L ≈ 576 nm, d = 10 nm, n = 10^{12} cm^{-2}. M_th is calibrated in situ by approach to self-oscillation;

the bias is then set at the tracked ratio M/M_th = 0.7. Whether the cell self-oscillates at 353 K is itself open at the kinetic tier — the kinetic operator specified below, run with a single relaxation rate, finds no threshold there for A ≤ 1 when the electrons shed their drift heat, and at 300 K one only at A = 3 — so the calibration step is also a test (in-model; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md, discriminant D2). At 353 K an onset below a drift of 0.66 v_F (6.6×10^5 m/s) that oscillates at the cavity line contradicts the single-rate statement and is consistent with two of the three admissible two-rate continuations; an onset at another frequency is not a plasmon threshold; and a failure to self-oscillate up to the current the cell tolerates is consistent with the statement but does not discriminate against contact loss. The five gates are listed in Table 7.

Gate	Observable	Pass threshold	Model target
G1	single-cell per-pulse net gain at ≤ 80 °C	≥ +2.6 dB (beats the per-gate loss)	+8 dB (streaming, per-slot peak; a design-length rerun measures +9.2 dB — Section 7.3); drive pinned relative to the measured knee: source amplitude ≈0.76–1.15 %, −1 dB at intracavity swing ≈14 ± 2 %, operating drive ≈3.8× below the pulse knee; a promoted re-issue proposes, without amending this row, pinning the conventions as G1a (existence) / G1b (knee report), with +10.2 dB (CW) as its expectation for G1a, the +8 dB being the shipped configuration's streaming reading (Section 10)
G2	pulse-to-pulse gain spread at 0.2–0.25 THz	< 1 dB	< 0.1 dB for the clock train (data-independent); with PRBS data the in-model expectation at the 4-ps slot is ≈3–4 dB p-p (Section 6.2); at the gated fan-out-2 slots the released solver reads 0.04–0.31 dB (Section 10) — the

			pass threshold is unchanged
G3	cell noise figure	≤ 6 dB	≈ 3 dB (floor 2.77 dB at 353 K; below 3.01 dB at any gain within Eq. (7), Section 10)
G4	two cascaded cells, one junction	≥ 1.7× the single-cell decibel gain	≈ 2×
G5	thermal compliance at operating duty	ΔT within the 55 K budget	≈ 23 % margin at fill ⅓ (hydrodynamic tier; ≈ 4 % within the closure of Section 7.5; every computed row of graphene's own fluid with a threshold exceeds the budget under that fluid's own Joule form $J \cdot E_0$, in-model — Section 8.2)

Table 7. *The five pre-registered pass/fail gates. G1 is the existence proof of regenerative DS gain; G4 — the first concatenation of room-temperature plasmonic gain ever attempted — is the single most informative outcome, pass or fail.*

Community notes correct, or propose to pin, several of the conventions behind the model targets of Table 7; no pass threshold moves. For G1, notes/2026-08-01-bench-gate-g1-reissued.md proposes pinning the existence reading, G1a, to a retuned cavity, a CW drive and the intracavity-peak observable at a measurement drive ≥ 10× below the measured knee and a bias of 0.7 × the device's own measured threshold, where the proposal expects +10.2 dB (in-model; the +8 dB of the table is the shipped configuration's streaming reading, and the proposed CW pin would narrow the row's 'per-pulse' wording); a knee report, G1b — a report, not a gate — would add the drive sweep G(A) to 1-dB compression for the CW and the duty-0.8 gated drives (gated knee ≈1.2 % source amplitude at N = 240), measuring A_SAT in place of the imported A_op; and a G1a pass would not close fan-out 2, a comparison that stays with G4. For G2 the ≈3–4 dB expectation belongs to the 4-ps slot; at the gated slots of Section 6.2(iii) the released solver reads, for duty-0.8 launches, a raw '1'-level spread of 0.04–0.18 dB at the analytic bias and 0.05–0.31 dB at the ratio bias in the 67–71 GHz fan-out-2 window, and, at 91 GHz, a fan-out-1 slot, 0.29–0.43 dB at the analytic bias and 0.40–0.79 dB at the ratio bias, growing with grid refinement (demonstrated rows; notes/2026-07-31-physical-launch-gated-frontier.md, notes/2026-08-01-gated-requirement-round-trip.md, notes/2026-08-01-ratio-bias-gain-table.md); a re-issue of G2 for the gated slot is open. For G3, no gain takes the bare cell above 3.01 dB within Eq. (7), so a reading between that and the 6 dB threshold indicates noise added ahead of the gain or a hotter-than-matched idler (in-model, within the released amplifier rule; Section 7.4).

The die plan includes passive twins for differential measurement, junction variants (Section 9.4), a d = 5 nm variant ($s \propto \sqrt{d}$ in the classical limit; the model chain scores it as a

+2.6 dB-class gain lever at the hydrodynamic tier, though in the single-rate kinetic operator at 353 K, with A = 1 and the drift heat shed, it does not restore a threshold, Section 7.5), and a graphite-gated disorder de-risk die. The calibration sequence measures on-die every quantity this manuscript has had to assume, in order: the disorder σ, the lifetime τ(T), the plasmon speed s (fixing ε_z in Eq. (1)), the threshold M_th, the launch/read-out coupling efficiency, and the launcher's amplitude jitter at the launch plane (Section 9.5). Two of those measurements now carry kinetic freight (Section 7.5), and in both cases what the bench returns is broader than what the closure signs — so neither is pre-registered here as a kinetic expectation. The measured s is the physical speed, while Eq. (1) is the collisionless endpoint; fitting ε_z to the measured s therefore absorbs whatever kinetic shift is present into a dielectric constant, and the fitted ε_z should be reported as what it is rather than as a material property. On the size of that shift the model does not settle the bench value: the closure signs an in-closure deficit of −1.8 % at 353 K, while the single-rate kinetic operator, which carries the finite-temperature part the closure leaves uncomputed, puts the zero-drift kinetic cavity line 9.6–13.4 % below the released model's continuum line, outside the closure's own [−4, +1] % (in-model; Section 7.5), so the total shift the bench measures is open. What survives regardless is a requirement, not a prediction: the cavity is fixed-length, a resonance offset of this order costs gain (Section 7.5), and the density trim that nulls it — $\Delta n/n \approx +7$ % if the in-closure deficit is what the bench finds, ≈ +48 % if the kinetic bulk's is — must be sized as a capability and commissioned against the measured offset. G1 is where it is first measurable. The measured M_th, likewise, is the physical threshold, if the cell has one (Section 7.5), rather than the analytic 0.147083, and the closure expects it higher — ≈ 0.165 at 353 K, ×1.12, in-model within that closure. That figure is not a bench band. It is quoted for the specular-clamp solve of the paragraph below, whereas a bench-measured threshold absorbs real contact non-ideality by construction (Section 9.1) — the same additive boundary channel the band holds open at ±1 dB, and the diffuse-contact sweep may move the prediction further. One caution attaches either way: ≈ 0.165 lands numerically near the retired numerical-diffusion artifact of Section 7.3 (0.169). The proximity is a coincidence, and the two must not be conflated or double-counted.

The remaining modeling tier — a Boltzmann–Maxwell grating-gate solve, with method precedent in [18] — owns five questions with fully specified inputs: the regenerative gain curve G(M/M_th) under kinetic and electromagnetic corrections (the tier's highest priority, and now a falsifiable one: the kinetic part of the answer must land inside the signed band of Section 7.5); real junction loss; radiative coupling between cells (the parasitic channel Section 7.5 leaves to this tier; its charge-channel exclusion holds only under a closed-form ansatz); absolute noise calibration against the 2.77 dB (353 K) floor; and the launch-transducer coupling efficiency. The promoted kinetic note (Section 7.5) turns the first of those from a topic into a specification, and the tier's scoping should follow it. The accuracy target is set by the band the solve must adjudicate: to be decisive against [−1.5, −0.3] dB it must resolve per-cell gain to ±0.25 dB — the round-trip amplitude loop to ±0.0024 about its analytic 0.916 baseline, per-unit-time damping to $\pm 5\times10^{9}$ s^{-1}, and additive boundary dissipation to under 1 % per round trip. For adjudicating that band the ingredients then rank, and not in the order the tier's name suggests; whether a 353 K threshold exists at all comes before the band and is owned by collision-operator fidelity (below). Contact and boundary-condition fidelity comes first: it owns the only unbounded allowance (±1 dB, sign

open, −0.28 dB per 1 % of additive round-trip loss), because a Knudsen layer of ℓ_ee ≈ 83 nm sits at each clamp at the 80 °C cap, ≈ 14 % of the cell; the multiplicative part of the same physics is tracking-protected to under 0.05 dB and needs nothing from the solve. The boundary channel also now has a cheap bench test ahead of the solve: because δ_b is a per-reflection loss while the bulk viscous channel is distributed, three sub-threshold cavity-linewidth signatures separate them — a contact-preparation contrast isolates δ_b directly (≈3.2 GHz of excess half-linewidth per 1 % additive round-trip loss), a temperature sweep reads its Knudsen exponent (the bulk viscous term peaks at ωτ_ee ≈ 1 near 255 K while δ_b is monotone in T), and a drift sweep reads the sign of the drain term (notes/2026-07-18-boundary-channel-bench-discriminants.md; posted as a measurement request, issue #88, and as Stage 0 of the staged bench plan's open call for an experimental partner, issue #97). A near-model linewidth on a specular-contact cell therefore gates an architecture-led Part III on data, without waiting for the full kinetic solve; a broad line measures the three inputs the solve otherwise owes. Part III carries this measurement as its prerequisite gate L0, unmade (Part III, Section 2.4 and Table T10 [45]). The same sub-threshold line also reads graphene's electron fluid. Between 300 and 353 K at fixed n and cell length, the zero-drift kinetic cavity line falls 2.8–3.5 % in the single-rate kinetic operator, against 0.12 % for the released model's continuum line (and none for Eq. (1)): a fall above 2 % separates the two and one below 0.5 % favours the released model, although under slow relaxation of the current not locked to momentum the fall is only ≈ 1.8 %, so the sweep reads that channel's rate as much as the temperature. The zero-drift line's width and shift, read together, bracket that open rate once δ_b is separated by the contact-preparation contrast, with τ from DC transport and the viscosity measured independently, and with the ε_z calibration above inherited (in-model; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md, discriminants D1 and D4). Both are discriminants rather than falsifiers, under the G1a protocol that notes/2026-08-01-bench-gate-g1-reissued.md proposes, and both depart from its proposed retuning pin: the carrier is not re-trimmed between temperatures. Collision-operator fidelity at ωτ_ee ≈ 0.5–1 comes second: it owns the width of the signed band, and pinning the stress-relaxation time to within a factor of two — by computation, or by measuring the mode linewidth on the bench — shrinks that width to about ±0.3 dB. It also owns more than the width. In the relaxation-time operator specified below, run with a single rate in the finite-temperature bulk, about two-thirds of the A = 1 damping at 353 K is the electron-electron relaxation of the current not locked to momentum (12.9 % of the Drude weight), a rate the stress-relaxation band does not bound: over two-rate variants in which that rate runs from 10^{-3} to 10^{3} times the stress-relaxation rate (the single-rate operator sets them equal) it sets the damping between 0.118 and 0.458 of $1/(2\tau)$, so the linewidth alone is ambiguous in it. Whether a 353 K threshold exists at all turns on that rate and on the operator's finite-drift structure, which relaxation-time models do not pin — three admissible continuations, identical at zero drift, give three answers (in-model; the rate is open; notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md). A collision-integral computation at T/T_F = 0.26 would pin it, and on the bench the zero-drift line's width and shift bracket it (above). Any collision operator the kinetic solve uses must pass that note's four admissibility gates: dissipation in the entropy metric, self-adjointness in that metric, no sourceless bulk growth, and a stable driven base. Electromagnetic coupling comes third, and is not required for this question at all: the cell is strongly gated ($q_0 d = 0.027$) and quasistatic ($\omega/cq_0 = 8\times10^{-3}$, a 10^{-4} effect in

the dispersion), so the full-wave tier owns the other questions on the list — radiative parasitics, launch efficiency, absolute noise — and not this one. The recommendation carried here is therefore to split the tier's first deliverable in two: a kinetic cavity solve — linearized Boltzmann on the 2D Dirac cone with a number-, momentum- and energy-conserving relaxation-time operator (its rate for the current not locked to momentum and its finite-drift structure stated, since existence turns on both) about the drifted local equilibrium, on the 1D gated cavity with its quasistatic Poisson kernel, the source density clamp and drain current clamp imposed at the kinetic level, a specular-to-diffuse contact parameter swept $0 \to 1$, reporting $\omega''(M)$, the kinetic threshold, and the driven transfer at $0.7 \cdot M_{th,kin}$ — and, separately, the electromagnetic tier. The kinetic solve is a workstation-days job at $\sim 10^3$ spatial by $\sim 10^2$ angular resolution, not a cluster job, and it is pre-registered against four falsifiable quantities: a tracked gain correction of −0.95 dB in [−1.5, −0.3] at 353 K; −1.27 dB in [−1.6, −0.5] at 300 K; a kinetic threshold of 0.165 in [0.154, 0.169] with specular clamps, the diffuse-contact sweep being free to move it further; and the in-closure resonance shift of Section 7.5, −1.8 % in [−2.3, −1.2] % at 353 K. These are targets for the solve, not for the bench, whose measurements carry channels the solve holds fixed (Section 9.1). A result outside those bands falsifies the closure of Section 7.5 — not the released chain, whose outputs none of them contest. The operator specified above has also been run in a simpler setting — a single relaxation rate in the bulk, with the DS clamps on the plasmon pair rather than at the kinetic level — so that run is not this solve. It leaves the 353 K gain correction and kinetic threshold without an operating point — when the electrons shed their drift heat no threshold exists for $A \le 1$ below the Landau ceiling, so the tracked correction would be undefined rather than shifted, and at 300 K a tracked point exists only at $A = 3$ — while the closure, fed to the same cavity, returns 0.165916, inside its band (in-model; Section 7.5). The solve is therefore asked first whether a threshold exists. Under the falsifier above, a solve that finds no 353 K threshold returns no value inside the closure's threshold and gain bands, and so counts against the closure rather than as a failure to land.

One open item from the abstract's list does not appear above because it is not on the bench's critical path: the photonic synthesis of the pulsed clock (Section 3.2). The bench drives its cells through the photoconductive launch chain of Section 3.3; the multi-line comb extension is what the full fabric — not the experiment — waits on, together with the choice of source class and the clock's distribution budget, open design questions tracked as issue #139 (Section 3.2).

11. Conclusion and outlook

The design's feasibility rests on a single stated physics question — can room-temperature graphene deliver net plasmon gain at ~1 THz? Five passes of reduced-order modeling answer it as well as reduced-order physics can: yes in-model, by operating a resonant DS cell as a sub-threshold regenerative amplifier, at a density chosen by disorder rather than heat (on the design model's sharpness, Section 7.4), clocked optically because electrical clocking was never physically available, cascaded without gain degradation on the cascade budget of Section 7.4, near the thermal noise floor. The logic family built on that amplifier is the design model's: the released cell compresses rather than thresholds and supplies no density route to a NOT (Section 5).

The answer is bounded in four ways, each stated where it arises: the experimental cornerstone [2] is cited for what its authors claim — current-pumped plasmonic amplification — not for what the design needs (DS-cavity gain, which no experiment has shown); signals route with the electron drift, and loop suppression is architectural rather than free; the operating point exists only as a tracked threshold ratio, never as an absolute bias; and the hydrodynamic certification at 300 K is marginal, which is why the kinetic solve leads the remaining-work list. Within the closure of Section 7.5 that fourth boundary has a sign, and the sign is negative: about a decibel off every cell at the 80 °C cap and rather more at 300 K, in-model, with an open ±1 dB allowance at the contacts riding on top of it (Section 7.5). None of these boundaries has killed the design in-model, but the fourth could: it leaves open whether the 353 K operating point exists at the kinetic tier at all, since the single-rate kinetic operator finds no 353 K threshold to track for $A \leq 1$ when the electrons shed their drift heat, and whether graphene's own electron fluid has one is open (Section 7.5). The nearest thing to a casualty is the thermal margin, which the same correction reads down from 23 % to a few percent within the closure, and which every computed row of graphene's own fluid that has a threshold overruns under that fluid's own Joule form $J \cdot E_0$ (Section 8.2). Each narrows what must be believed without measurement — and the kinetic one narrows it in a falsifiable direction: the next tier's solve will land inside the pre-registered bands or break them, and a solve that finds no threshold to track at 353 K counts as breaking them (Section 10).

None of the in-model statements is yet a measurement. The design should be judged by the protocol gates of Section 10, and the most informative outcome — pass or fail — is G4: two cells, one junction, the first concatenation of room-temperature plasmonic gain ever attempted. It is the same property that lets the half-adder block of Section 6.5 drive its successor — the property that makes five cells a computer rather than a demonstration. The model chain is exhausted; the design is now exactly as falsifiable as it was always meant to become.

Appendix A. The model chain: conventions and the released code

The five passes are the manuscript's entire evidentiary basis, and they are released as a small, runnable Python package (Table A1; depends only on NumPy, and Matplotlib for the figures). One command, run_all.py, reproduces every number in Sections 7–8 and writes them to a results file; figures.py regenerates Figures 6–10. The shared conventions are: damping enters at the amplitude convention, ω'' compared against $1/(2\tau)$ (Eq. 2–3); "per-gate loss" means one $\lambda_p/2$ of propagation (2.17 dB at 300 K, 2.56 dB at 353 K at the design Q) — the quarter-wave cell's single transit is half that; all temperatures are tagged, with $\tau(T) = \tau(300\ \mathrm{K}) \cdot (300/T)$; solver bias values are reported as ratios to the solver's own threshold on the grid run, except in rows labelled as run at the analytic bias (0.7 of Eq. (3)'s threshold) or at an absolute bias (a fixed M, 0.7 of the N = 240 threshold, held at every grid; it equals the ratio only at N = 240); at the shipped grid that is the coarse-grid threshold ($M_{th,num} \approx 0.169$, vs analytic 0.147 at 353 K — a Lax-Friedrichs numerical-diffusion artifact whose $\Delta x \to 0$ limit is the exact inviscid threshold ≈0.1493, not the analytic value; the refinement rerun at N = 480 measured 0.158748, confirming the pre-registered 0.1590

± 0.0010, and the kinetic threshold within the closure of Section 7.5 sits at 0.165 [0.154, 0.169]).

The solver's own threshold is a stated convention. M_th,num is the released estimator's root at N = 240, cfl = 0.4, seed 1 and 90 round trips, with a 40-bin fit over the last 60 % of the trace. cfl is the solver's time-step setting, not the Courant number: the step carries a 0.2 speed pad (Δt = cfl·Δx/cmax, cmax = 1 + |M| + 0.2), giving a true Courant number of 0.292 at the shipped point (notes/2026-08-07-mth-num-excess-decomposition.md). The shipped value is a root only at its own record length and fit convention — its eight-seed range at 90 round trips is 2.83×10^{-4}, and at every sampled record length ≥ 120 the growth rate at the shipped root is positive at 8 of 8 seeds (demonstrated; notes/2026-08-06-mth-num-record-length-and-fit-convention.md) — while the N = 480 confirmation holds on the eight-seed census, whose median is 0.15883 at both 90 and 240 round trips; from a grid in (860, 920] on, the released scan returns no value (Section 7.3). The time step is part of the convention: every solver figure is quoted at the released default; along the released refinement path Δt refines with Δx, so an order read there — as for the gated '0'-slot phase walk — does not separate the two (notes/2026-08-12-gated-zero-residual-phasor.md); and at fixed grid the quantities the scheme's dissipation and dispersion set move with cfl. The threshold's eight-seed median at N = 240 runs 0.1912 / 0.1762 / 0.1688 / 0.1643 at cfl 0.2 / 0.3 / 0.4 / 0.5, against −7.80 % for the whole 240 → 720 grid ladder (demonstrated; notes/2026-08-07-mth-num-excess-decomposition.md), and at the gated ratio bias (N = 240, 67 GHz) the ring-down decay moves by 14.7 dB per slot and the '0' residue after a single '1' by 10.3 dB across cfl 0.2–0.8, the released value being the optimistic corner in amplitude (demonstrated; notes/2026-08-12-gated-zero-residual-phasor.md).

Two properties of the chain are stated plainly so the "in-model" label is never mistaken for more than it is. The analytic backbone (Eqs. 1–7, the thermal and kinetic estimates) is closed-form and reproduces the operating point from first principles. The regenerative gain is owned by the analytic cavity transfer (regen.py) and independently confirmed by the nonlinear shallow-water solver (solver.py); the solver's absolute gain calibration carries the numerical-diffusion caveat above, along with the calibration structure now derived exactly by community review (July 2026): the exact linearized operator of the released scheme (runnable listing in the promoted notes) attributes 0.956 dB of the CW/streaming difference to linear numerics — drive detuning from the zero-drift cavity length, interior diffusion, and a drift-induced boundary loss — and the remaining 0.908 dB to pulse-train structure, a drive amplitude (3×10^{-3}) near the compression knee, and the per-slot estimator — a drive sweep since splits these as ≈0.06 dB compression and ≈0.87 dB estimator convention — which is why the per-pulse figure is quoted conservatively; the retuned-run and N = 480 predictions were run and confirmed in July 2026 (+9.22 dB measured; 0.158748 measured), with baselines replicated bit-for-bit — see notes/2026-07-12-predictions-resolved.md and REPLICATIONS.md. Two transfer-model parameters are stated explicitly: the cascade operating swing A_op = 0.0116 and the compression knee A_SAT = 0.02 (cell.py). The cascade figures of Section 7.4 (+8.4 to +9.3 dB per cell, ≈ 1.3 dB of compression at the operating swing) are evaluated at A_op, a released default whose origin is undocumented (open), though it is consistent with the measured compression curve to 0.24 dB (notes/2026-07-17-drive-sweep-knee-anchored.md). The fan-out-2

requirement ledger behind Section 6.2(iii) has the same constant as its one free parameter (demonstrated arithmetic; notes/2026-08-01-gated-requirement-round-trip.md), and that note recommends retiring it as a ledger input in favour of a measured drive sweep (the G1b report, Section 10). Eq. (5) itself is coded with A_in as the prefactor and has no caller in the released pipeline (Section 5). Table A1 maps each pass to its module:

Pass	Model	Released module(s)
1	drift-Doppler cascade + 1-D periodic-slab thermal solution	thermal.py, ds_cell.py — v_sat(n) = ω_OP/k_F, R_th, duty ceilings
2	DS increment, Eq. (2); break-even analytics; butterfly margins; ISI	ds_cell.py, cell.py — per-gate loss, k ≥ 8 design rule (assumed, not derived), gated ISI
3	nonlinear 1-D shallow-water DS solver + analytic cavity transfer	solver.py, regen.py — M_th,num, CW & pulse gain, saturation
4	cascade budget (compression form at A_op; no multi-cell cascade run); noise floor F = 2 − 1/G; disorder Monte-Carlo	cell.py, noise.py, disorder.py — per-cell gain, NF, swing-to-puddle yield on the design premise k = 8
5	coupled two-cell stability under a closed-form clamp ansatz; kinetic-validity estimates	coupled.py, kinetic.py — clamp-stability, τ_ee, ωτ_ee, viscosity

Table A1. *The released model chain, pass by pass. Each module is independently runnable and self-checks against the manuscript values; run_all.py consolidates them. The package is archived with the manuscript under a persistent DOI and linked from the Data and Code Availability statement.*

Data and code availability. The five-pass model chain is released as a runnable Python package (Appendix A) and is archived with the manuscript at https://github.com/ryoji-info/FableComputer; run_all.py reproduces every quantitative claim in Sections 7–8 and figures.py regenerates Figures 6–10. The same repository hosts the community's technical notes (notes/), which together with the released chains make up the community record; the notes promoted through 17 October 2026 that bear on this paper are incorporated in its text (papers/REVISION-HISTORY.md lists them). Items tracked as repository issues (https://github.com/ryoji-info/FableComputer/issues) are cited as open design questions or requests, not as results: the linewidth measurement request (issue #88), the call for an experimental partner (issue #97), and the clock source, the material gap and the CMOS boundary (issues #139–#141). The complete revision history of the manuscripts is kept in papers/REVISION-HISTORY.md; corrections adopted by the community record but not yet in a deposited revision are indexed in papers/ERRATA.md.

Appendix B. Symbols

Table B1 collects the symbols used throughout the manuscript, with their values at the operating point.

Symbol	Meaning	Value at the operating point
f_0, ω	carrier frequency, angular frequency	~1 THz, $2\pi\times10^{12}$ s^{-1}

τ, Q	plasmon momentum-relaxation time; Q = ωτ	1.0 ps / 6.3 (300 K); 0.85 ps / 5.3 (353 K)
n, E_F	carrier density; Fermi level	10^{12} cm^{-2}; 117 meV
s, λ_p	acoustic-plasmon speed, Eq. (1); wavelength s/f_0	2.2–2.3×10^6 m/s (design 2.33×10^6); ≈ 2.3 μm
d, ε_z	gate–graphene hBN thickness; its out-of-plane permittivity	10 nm; ≈ 3.4–3.8
L	cell length, Eq. (4)	≈ 576 nm
v_0, M	drift velocity; Mach number v_0/s	≈ 2.4×10^5 m/s; ≈ 0.10 (at 353 K)
M_th	DS threshold, Eq. (3); tracked in situ	0.147 (353 K), 0.125 (300 K); coarse-grid solver 0.169 (N = 240, cfl = 0.4, seed 1, 90 round trips; Δx→0 → exact inviscid ≈0.1493; kinetic 0.165 within the Section 7.5 closure)
g(n), n_th, Δn, β, ℓ	cell transfer model of the design, Eq. (5)	design model; the released cell's own g(n) is a resonance curve (Section 5)
A_pump, A_in, A_swing	clock-pulse and logic-input amplitudes, Eq. (5); rail-to-rail input swing (Section 5)	—
κ±	spatial attenuation of co-/counter-drift waves, Eq. (6)	contrast ≈ 0.26–0.31 dB per cell transit
k	threshold sharpness β·A_swing/Δn	≥ 8 (design rule; the released cell's density coupling gives 0.18–0.26, Section 5)
G, F	cell gain; noise figure, Eq. (7)	+8 dB/pulse streaming (CW +9.7 dB); floor 2.77 dB (353 K)
σ	charge-puddle density spread	2.5×10^9–4×10^{10} cm^{-2} (literature)
τ_ee	electron-electron collision time	≈ 115 fs (300 K) ⇒ ωτ_ee ≈ 0.7
A	multiplier on the electron-electron rate estimate of [15] (Sections 5, 7.5)	1 (admissible band 0.5–3)
J	drift current density n·e·v_0	≈ 0.38 mA/μm

Table B1. *Nomenclature. Each value is quoted at the operating point of Table 3; the section where it is derived is cited in the text.*

References

[1] R. Furui, “A 100-GHz Room-Temperature Half Adder Using Graphene Plasmons as Bits, Driven by a Microcomb-Photomixing Carrier: Architecture, Quantitative Roadmap, and Machine-Learning-Assisted Design Optimisation,” self-published manuscript, https://github.com/r-coin/basic/blob/master/hardware%20solutions/THz_Half_Adder_100GHz_RT.pdf

[2] S. Boubanga-Tombet, W. Knap, D. Yadav, A. Satou, D. B. But, V. V. Popov, I. V. Gorbenko, V. Yu. Kachorovskii, and T. Otsuji, “Room-Temperature Amplification of Terahertz Radiation by Grating-Gate Graphene Structures,” Phys. Rev. X 10, 031004 (2020). doi:10.1103/PhysRevX.10.031004

[3] V. Ryzhii, M. Ryzhii, and T. Otsuji, “Negative dynamic conductivity of graphene with optical pumping,” J. Appl. Phys. 101, 083114 (2007). doi:10.1063/1.2717566

[4] K. Yoshioka, G. Bernard, T. Wakamura, M. Hashisaka, K. Sasaki, S. Sasaki, K. Watanabe, T. Taniguchi, and N. Kumada, “On-chip transfer of ultrashort graphene plasmon wave packets using terahertz electronics,” Nat. Electron. 7, 537–544 (2024). doi:10.1038/s41928-024-01197-x

[5] Y. Tokizane, H. Kishikawa, T. Kikuhara, M. Talara, Y. Makimoto, K. Yamaji, Y. Okamura, K. Nishimoto, E. Hase, I. Morohashi, A. Kanno, S. Hisatake, N. Kuse, T. Nagatsuma, and T. Yasui, “Beyond 350 GHz: Single-channel 112 Gbps photonic wireless transmission at 560 GHz using soliton microcombs,” Commun. Eng. 5, 77 (2026). doi:10.1038/s44172-026-00659-8

[6] K. Hirota, T. Kashiwazaki, G. Ha, T. Yamashima, P. Jaturaphagorn, T. Suzuki, K. Takahashi, A. Kawasaki, A. Inoue, W. Asavanant, M. Endo, T. Umeki, and A. Furusawa, “Generation of 10-dB squeezed light from a broadband waveguide optical parametric amplifier with improved phase locking method,” Opt. Express 34, 7958–7972 (2026). doi:10.1364/OE.585323

[7] K. Kusyak, B. Schulte, T. Matsuyama, G. Kipp, H. M. Bretscher, M. W. Day, G. Meier, A. M. Potts, and J. W. McIver, “Monolithic optoelectronic circuit design for on-chip terahertz applications,” APL Photonics 10, 076117 (2025). doi:10.1063/5.0278183

[8] M. A. Haque, M. S. Akhter, I. Das, T. U. Zaman, J.-J. Tiang, N. S. S. Singh, A. D. Algarni, and A. A. Ateya, “Graphene based terahertz MIMO antenna with machine learning regression for 6G communications,” Sci. Rep. 16, 3900 (2026). doi:10.1038/s41598-025-32487-9

[9] V. E. Dorgan, M.-H. Bae, and E. Pop, “Mobility and saturation velocity in graphene on SiO_2,” Appl. Phys. Lett. 97, 082112 (2010). doi:10.1063/1.3483130

[10] D. Barcons Ruiz, N. C. H. Hesp, H. Herzig Sheinfux, C. Ramos Marimón, C. M. Maissen, A. Principi, R. Asgari, T. Taniguchi, K. Watanabe, M. Polini, R. Hillenbrand, I. Torre, and F. H. L. Koppens, “Experimental signatures of the transition from acoustic plasmon to electronic sound in graphene,” Sci. Adv. 9, eadi0415 (2023). doi:10.1126/sciadv.adi0415

[11] M. Dyakonov and M. Shur, “Shallow water analogy for a ballistic field effect transistor: New mechanism of plasma wave generation by dc current,” Phys. Rev. Lett. 71, 2465–2468 (1993). doi:10.1103/PhysRevLett.71.2465

[12] J. Xue, J. Sanchez-Yamagishi, D. Bulmash, P. Jacquod, A. Deshpande, K. Watanabe, T. Taniguchi, P. Jarillo-Herrero, and B. J. LeRoy, “Scanning tunnelling microscopy and spectroscopy of ultra-flat graphene on hexagonal boron nitride,” Nat. Mater. 10, 282–285 (2011). doi:10.1038/nmat2968

[13] K. Nagashio, T. Nishimura, and A. Toriumi, “Estimation of residual carrier density near the Dirac point in graphene through quantum capacitance measurement,” Appl. Phys. Lett. 102, 173507 (2013). doi:10.1063/1.4804430 (SiO_2-substrate measurement; cited as the disorder upper bound)

[14] P. Huang, E. Riccardi, S. Messelot, H. Graef, F. Valmorra, J. Tignon, T. Taniguchi, K. Watanabe, S. Dhillon, B. Plaçais, R. Ferreira, and J. Mangeney, “Ultra-long carrier lifetime in neutral graphene-hBN van der Waals heterostructures under mid-infrared illumination,” Nat. Commun. 11, 863 (2020). doi:10.1038/s41467-020-14714-1

[15] A. Lucas and K. C. Fong, “Hydrodynamics of electrons in graphene,” J. Phys.: Condens. Matter 30, 053001 (2018). doi:10.1088/1361-648X/aaa274

[16] D. A. B. Miller, “Attojoule optoelectronics for low-energy information processing and communications,” J. Lightwave Technol. 35(3), 346–396 (2017). doi:10.1109/JLT.2017.2647779

[17] IEEE, “International Roadmap for Devices and Systems (IRDS), 2023 Update,” IEEE (2023). https://irds.ieee.org/editions/2023

[18] Z. Qiu, S. Yang, S. Huang, S. Wang, P. Zhang, and Y. Gong, “Terahertz Amplification Induced by Electron–Phonon Interactions in Gated Graphene Plasmonic System,” Research 8, 1023 (2025). doi:10.34133/research.1023

[19] J. Crabb, X. Cantos-Roman, J. M. Jornet, and G. R. Aizin, “Hydrodynamic theory of the Dyakonov–Shur instability in graphene transistors,” Phys. Rev. B 104, 155440 (2021). doi:10.1103/PhysRevB.104.155440

[20] A. S. Petrov, D. Svintsov, V. Ryzhii, and M. S. Shur, “Amplified-reflection plasmon instabilities in grating-gate plasmonic crystals,” Phys. Rev. B 95, 045405 (2017). doi:10.1103/PhysRevB.95.045405

[21] Y. Koseki, V. Ryzhii, T. Otsuji, V. V. Popov, and A. Satou, “Giant plasmon instability in a dual-grating-gate graphene field-effect transistor,” Phys. Rev. B 93, 245408 (2016). doi:10.1103/PhysRevB.93.245408

[22] G. R. Aizin, J. Mikalopas, and M. S. Shur, “Plasma instability and amplified mode switching effect in THz field effect transistors with a grating gate,” Phys. Rev. B 107, 245424 (2023). doi:10.1103/PhysRevB.107.245424

[23] D. Veksler, F. Teppe, A. P. Dmitriev, V. Yu. Kachorovskii, W. Knap, and M. S. Shur, "Detection of terahertz radiation in gated two-dimensional structures governed by dc current," Phys. Rev. B 73, 125328 (2006). doi:10.1103/PhysRevB.73.125328

[24] Y. Dong, L. Xiong, I. Y. Phinney, Z. Sun, R. Jing, A. S. McLeod, S. Zhang, S. Liu, F. L. Ruta, H. Gao, Z. Dong, R. Pan, J. H. Edgar, P. Jarillo-Herrero, L. S. Levitov, A. J. Millis, M. M. Fogler, D. A. Bandurin, and D. N. Basov, "Fizeau drag in graphene plasmonics," Nature 594, 513–516 (2021). doi:10.1038/s41586-021-03640-x

[25] W. Zhao, S. Zhao, H. Li, S. Wang, S. Wang, M. I. B. Utama, S. Kahn, Y. Jiang, X. Xiao, S. Yoo, K. Watanabe, T. Taniguchi, A. Zettl, and F. Wang, "Efficient Fizeau drag from Dirac electrons in monolayer graphene," Nature 594, 517–521 (2021). doi:10.1038/s41586-021-03574-4

[26] T. A. Morgado and M. G. Silveirinha, "Directional dependence of the plasmonic gain and nonreciprocity in drift-current biased graphene," Nanophotonics 11, 4929–4936 (2022). doi:10.1515/nanoph-2022-0451

[27] D. Correas-Serrano and J. S. Gomez-Diaz, "Nonreciprocal and collimated surface plasmons in drift-biased graphene metasurfaces," Phys. Rev. B 100, 081410(R) (2019). doi:10.1103/PhysRevB.100.081410

[28] X. Mei, W. Yoshida, M. Lange, J. Lee, J. Zhou, P.-H. Liu, K. Leong, A. Zamora, J. Padilla, S. Sarkozy, R. Lai, and W. R. Deal, "First Demonstration of Amplification at 1 THz Using 25-nm InP High Electron Mobility Transistor Process," IEEE Electron Device Lett. 36, 327–329 (2015). doi:10.1109/LED.2015.2407193

[29] T. Maekawa, H. Kanaya, S. Suzuki, and M. Asada, "Oscillation up to 1.92 THz in resonant tunneling diode by reduced conduction loss," Appl. Phys. Express 9, 024101 (2016). doi:10.7567/APEX.9.024101

[30] H. A. Hafez, S. Kovalev, J.-C. Deinert, Z. Mics, B. Green, N. Awari, M. Chen, S. Germanskiy, U. Lehnert, J. Teichert, Z. Wang, K.-J. Tielrooij, Z. Liu, Z. Chen, A. Narita, K. Müllen, M. Bonn, M. Gensch, and D. Turchinovich, "Extremely efficient terahertz high-harmonic generation in graphene by hot Dirac fermions," Nature 561, 507–511 (2018). doi:10.1038/s41586-018-0508-1

[31] K. K. Likharev and V. K. Semenov, "RSFQ logic/memory family: a new Josephson-junction technology for sub-terahertz-clock-frequency digital systems," IEEE Trans. Appl. Supercond. 1, 3–28 (1991). doi:10.1109/77.80745

[32] A. V. Chumak, V. I. Vasyuchka, A. A. Serga, and B. Hillebrands, "Magnon spintronics," Nat. Phys. 11, 453–461 (2015). doi:10.1038/nphys3347

[33] L. Wang, I. Meric, P. Y. Huang, Q. Gao, Y. Gao, H. Tran, T. Taniguchi, K. Watanabe, L. M. Campos, D. A. Muller, J. Guo, P. Kim, J. Hone, K. L. Shepard, and C. R. Dean, "One-dimensional electrical contact to a two-dimensional material," Science 342, 614–617 (2013). doi:10.1126/science.1244358

[34] S. Zhang, J. M. Silver, X. Shang, L. Del Bino, N. M. Ridler, and P. Del'Haye, "Terahertz wave generation using a soliton microcomb," Opt. Express 27, 35257–35266 (2019). doi:10.1364/OE.27.035257

[35] S. T. Cundiff and A. M. Weiner, "Optical arbitrary waveform generation," Nat. Photonics 4, 760–766 (2010). doi:10.1038/nphoton.2010.196

[36] E. Rouvalis, C. C. Renaud, D. G. Moodie, M. J. Robertson, and A. J. Seeds, "Traveling-wave uni-traveling carrier photodiodes for continuous wave THz generation," Opt. Express 18, 11105–11110 (2010). doi:10.1364/OE.18.011105

[37] H. Ito, T. Furuta, F. Nakajima, K. Yoshino, and T. Ishibashi, "Photonic generation of continuous THz wave using uni-traveling-carrier photodiode," J. Lightwave Technol. 23, 4016–4021 (2005). doi:10.1109/JLT.2005.858221

[38] T. Ishibashi and H. Ito, "Uni-traveling-carrier photodiodes," J. Appl. Phys. 127, 031101 (2020). doi:10.1063/1.5128444

[39] P. Alonso-González, A. Y. Nikitin, Y. Gao, A. Woessner, M. B. Lundeberg, A. Principi, N. Forcellini, W. Yan, S. Vélez, A. J. Huber, K. Watanabe, T. Taniguchi, F. Casanova, L. E. Hueso, M. Polini, J. Hone, F. H. L. Koppens, and R. Hillenbrand, "Acoustic terahertz graphene plasmons revealed by photocurrent nanoscopy," Nat. Nanotechnol. 12, 31–35 (2017). doi:10.1038/nnano.2016.185

[40] M. B. Lundeberg, Y. Gao, R. Asgari, C. Tan, B. Van Duppen, M. Autore, P. Alonso-González, A. Woessner, K. Watanabe, T. Taniguchi, R. Hillenbrand, J. Hone, M. Polini, and F. H. L. Koppens, "Tuning quantum nonlocal effects in graphene plasmonics," Science 357, 187–191 (2017). doi:10.1126/science.aan2735

[41] A. Woessner, M. B. Lundeberg, Y. Gao, A. Principi, P. Alonso-González, M. Carrega, K. Watanabe, T. Taniguchi, G. Vignale, M. Polini, J. Hone, R. Hillenbrand, and F. H. L. Koppens, "Highly confined low-loss plasmons in graphene–boron nitride heterostructures," Nat. Mater. 14, 421–425 (2015). doi:10.1038/nmat4169

[42] G. X. Ni, A. S. McLeod, Z. Sun, L. Wang, L. Xiong, K. W. Post, S. S. Sunku, B.-Y. Jiang, J. Hone, C. R. Dean, M. M. Fogler, and D. N. Basov, "Fundamental limits to graphene plasmonics," Nature 557, 530–533 (2018). doi:10.1038/s41586-018-0136-9

[43] A. H. Barajas-Aguilar, J. Zion, I. Sequeira, A. Z. Barabas, T. Taniguchi, K. Watanabe, E. B. Barrett, T. Scaffidi, and J. D. Sanchez-Yamagishi, "Electrically driven amplification of terahertz acoustic waves in graphene," Nat. Commun. 15, 2550 (2024). doi:10.1038/s41467-024-46819-2

[44] Y. Dong, Z. Sun, I. Y. Phinney, D. Sun, T. I. Andersen, L. Xiong, Y. Shao, S. Zhang, A. Rikhter, S. Liu, P. Jarillo-Herrero, P. Kim, C. R. Dean, A. J. Millis, M. M. Fogler, D. A. Bandurin, and D. N. Basov, "Current-driven nonequilibrium electrodynamics in graphene revealed by nano-infrared imaging," Nat. Commun. 16, 3861 (2025). doi:10.1038/s41467-025-58953-6

[45] R. Furui, “The Fable Computer, Part III: A Ternary Language Model on the Regenerative Graphene-Plasmon Fabric,” v1.1, October 2026, not yet deposited; repository copy papers/Fable-Computer-Part-III.pdf, https://github.com/ryoji-info/FableComputer

[46] R. Furui, “The Fable Computer, Part II: Quantum-Limited Analog Tensor Processing on the Regenerative Graphene-Plasmon Fabric,” v2.0, July 2026. doi:10.5281/zenodo.21607070